

DIABETIC RETINOPATHY DIAGNOSIS AND CLASSIFICATION USING VISION TRANSFORMER

PROJECT REPORT – PHASE II

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DEPARTMENT OF INFORMATION TECHNOLOGY

BONAFIDE CERTIFICATE

This is to certify that the project work phase II entitled “**DIABETIC RETINOPATHY DIAGNOSIS AND CLASSIFICATION USING VISION TRANSFORMER**” is a bonafide work done by **HARIKANTH R [22UIT042]**, **SABARINATH A [22UIT125]**, **NAVEENKUMAR S [22UIT091]** in partial fulfilment of the requirements for the award of the B.Tech. Degree in Information Technology from Pondicherry University during the academic year 2025 – 2026.

PROJECT GUIDE

HEAD OF THE DEPARTMENT

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ABSTRACT

Diabetic Retinopathy (DR) is a progressive eye disease caused by long-term diabetes mellitus that damages retinal blood vessels and is a leading cause of vision loss among working-age individuals worldwide. Since the early stages often show no noticeable symptoms, delayed detection can lead to irreversible blindness, highlighting the need for reliable diagnostic systems. Traditional diagnosis through manual examination of retinal fundus images is time-consuming and depends heavily on expert interpretation. To address these challenges, this work proposes an automated DR detection and classification system using a Vision Transformer (ViT)-based deep learning framework. Unlike conventional Convolutional Neural Networks (CNNs) that mainly capture local features, the Vision Transformer utilizes a self-attention mechanism to learn both local and global relationships within retinal images. In the proposed approach, retinal images are preprocessed and divided into smaller patches that are transformed into embeddings and processed through transformer layers to extract meaningful spatial patterns. Based on these features, the model classifies retinal images into five stages of diabetic retinopathy: No DR, Mild, Moderate, Severe, and Proliferative DR. This approach improves the ability to identify complex retinal abnormalities and provides an efficient and scalable solution to assist healthcare professionals in early screening and diagnosis of diabetic retinopathy.

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LIST OF ABBREVIATIONS

ABBREVIATIONS	FULL FORM
XGBoost	– eXtreme Gradient Boosting
V-Net	– Volumetric Convolutional Neural Network
RNN	– Recurrent Neural Network
LSTM	– Long Short-Term Memory
DL	– Deep Learning
DNN	– Deep Neural Network
DQN	– Deep Q-Network
DT	– Decision Tree
DTD	– Deep Taylor Decomposition
DWT	– Discrete Wavelet Transform
ELM	– Extreme Learning Machine
ELMo	– Embeddings from Language Models
ELU	– Exponential Linear Unit
DR	– Diabetic Retinopathy
MA	– Microaneurysms
PCA	– Principal Component Analysis
CLAHE	– Contrast Limited Adaptive Histogram Equalization

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CHAPTER 1

INTRODUCTION

Diabetic Retinopathy (DR) is a severe microvascular complication of diabetes mellitus and one of the leading causes of preventable blindness worldwide. It develops when prolonged high blood glucose levels damage the retinal blood vessels, leading to vision impairment or complete loss if left untreated. According to the World Health Organization, the prevalence of diabetes is rapidly increasing, particularly in developing countries, which in turn elevates the risk of DR and other diabetes-related complications. Early detection of DR is crucial because the initial stages are often asymptomatic, and delayed diagnosis can result in irreversible retinal damage. Conventional screening methods rely on manual examination of retinal fundus images by ophthalmologists, which is not only time-consuming but also highly dependent on clinical expertise. The growing patient population presents a significant challenge for healthcare systems, emphasizing the need for scalable, efficient, and reliable diagnostic solutions.

Artificial Intelligence (AI) and deep learning have emerged as promising technologies to address these challenges, enabling automated analysis of medical images with high accuracy and efficiency. Convolutional Neural Networks (CNNs) have been widely used in DR detection due to their ability to extract local features and patterns from images. However, CNN-based approaches often face limitations in capturing long-range dependencies and understanding global contextual relationships across retinal images, which can reduce diagnostic accuracy and generalization. Vision Transformers (ViTs) provide a novel solution by dividing images into smaller patches and employing self-attention mechanisms to learn both local and global features simultaneously. This architecture allows the model to identify subtle pathological patterns across the entire retinal image, improving classification of DR severity while maintaining interpretability. By combining automated analysis with attention-based learning, ViTs enhance both the reliability and trustworthiness of AI-driven medical tools.

The DR diagnostic system integrates multiple stages, including image preprocessing, patch embedding, transformer-based feature extraction, classification, and explainable visualizations. Retinal fundus images are first enhanced to reduce noise and improve uniformity, then divided into patches and converted into numerical embeddings for the Vision Transformer. The model classifies images into five severity levels—No DR, Mild, Moderate, Severe, and Proliferative DR. Explainable AI (XAI) techniques generate attention maps to visually indicate regions..

1.1 PATHOPHYSIOLOGICAL FEATURES OF DIABETIC RETINOPATHY

Almost all individuals with diabetes will develop some degree of retinal damage, known as diabetic retinopathy, over the course of several decades. In many cases, early retinal changes are detectable only through a comprehensive retinal examination and may not cause noticeable vision problems. As the condition progresses, initial signs observed during an eye exam often include small bulges in the retinal blood vessels, called microaneurysms. Over time, more significant abnormalities can develop, such as cotton wool spots, hemorrhages, lipid deposits referred to as hard exudates, intraretinal microvascular anomalies, and abnormal retinal veins. In advanced stages, new blood vessels form across the retina, which are fragile and prone to bleeding. Minor bleeding can create floating dark spots that partially obstruct vision, while severe bleeding may cause complete vision loss.

Approximately 50% of individuals with diabetic retinopathy experience macular swelling, referred to as macular edema, which can occur at any stage of the disease. If the swelling affects the central part of the macula, it can lead to visual disturbances ranging from mild blurring to severe loss of central vision. Without treatment, around 30% of patients with macular edema may experience significant vision impairment within three to five years. Macular edema is the leading cause of vision loss in diabetic retinopathy, highlighting the importance of early detection and intervention.

The ongoing cycles of abnormal blood vessel growth, swelling, and scar formation can ultimately result in retinal detachment. This condition often presents as a sudden onset of floating dark spots, flashes of light, or blurred vision. Retinal detachment represents a serious complication of diabetic retinopathy and requires prompt medical attention to prevent permanent vision loss.

1.2 DIAGNOSIS AND CLASSIFICATION

Diabetic retinopathy is primarily diagnosed through retinal examination, typically using ophthalmoscopy or fundus imaging. According to the American Academy of Ophthalmology, diabetic retinopathy is classified into five progressive severity stages. The initial stage, “no apparent retinopathy,” represents a healthy retina with no detectable abnormalities. The subsequent stages—mild, moderate, and severe non-proliferative diabetic retinopathy (NPDR)—reflect escalating levels of retinal microvascular damage. Mild NPDR is characterized by the presence of microaneurysms without additional retinal pathology. Moderate NPDR demonstrates more extensive damage than mild NPDR but does not fulfill the

criteria for severe NPDR. Severe NPDR is identified by the presence of ≥ 20 retinal hemorrhages in each quadrant, venous beading in at least two quadrants, and prominent intraretinal microvascular abnormalities (IRMA) across the retina. The final stage, proliferative diabetic retinopathy (PDR), involves retinal neovascularization or pathological blood leakage into the vitreous cavity (vitreous hemorrhage) or between the vitreous and retina (preretinal hemorrhage).

Macular involvement is assessed separately and categorized based on the presence or absence of macular edema (ME). When present, macular edema is further classified as mild, moderate, or severe. Mild ME is defined by retinal thickening or lipid exudates located distant from the foveal center. Moderate ME involves thickening or deposits approaching the fovea, whereas severe ME affects the foveal center directly, posing a significant risk of central vision loss. Optical coherence tomography (OCT) is the preferred modality for quantitative assessment of macular edema, providing high-resolution cross-sectional images of retinal layers and fluid accumulation.

For a detailed evaluation of retinal microvasculature and precise localization of ischemic or leaky regions, fluorescein angiography (FA) is frequently employed by retina specialists. FA allows visualization of retinal perfusion, identification of microaneurysms, capillary nonperfusion, and neovascularization, assisting in both severity grading and therapeutic planning. These diagnostic tools collectively enable clinicians to monitor disease progression, stratify risk, and guide timely interventions to prevent vision-threatening complications.

1.3 ARTIFICIAL INTELLIGENCE

Artificial Intelligence (AI) refers to the simulation of human cognitive abilities in machines, enabling them to perform tasks that typically require human intelligence, such as learning, reasoning, and decision-making. In the context of diabetic retinopathy (DR), AI systems are designed to analyze retinal fundus images, detect subtle pathological features, and classify the severity of the disease. By mimicking the human visual assessment process, AI can assist ophthalmologists in screening large volumes of patients efficiently and accurately, reducing the risk of delayed or missed diagnoses. AI applications in DR leverage technologies such as machine learning, deep learning, and convolutional neural networks (CNNs), as well as advanced architectures like Vision Transformers (ViTs). These models are trained on large datasets of annotated retinal images to recognize microaneurysms, hemorrhages, exudates, and

other retinal abnormalities. Deep learning algorithms can automatically extract both local and global patterns from images, allowing for precise classification into stages ranging from mild non-proliferative DR to proliferative DR. The integration of Explainable AI (XAI) further enhances clinical trust by generating visual maps that highlight the regions influencing the model's decision.

By implementing AI-based DR detection systems, healthcare providers can achieve faster, more consistent, and scalable screening compared to traditional manual examination. Such systems not only assist ophthalmologists in early diagnosis but also help prioritize high-risk patients for timely intervention. The use of AI in DR screening exemplifies how intelligent systems can augment medical decision-making, improve patient outcomes, and reduce the burden of vision loss due to diabetes on a global scale.

TYPES OF ARTIFICIAL INTELLIGENCE

Artificial Intelligence (AI) is categorized into two main types: weak and strong.

- **WEAK AI:** This form of Artificial Intelligence embodies a system designed to carry out one particular job. Examples include video games (such as chess) and personal assistants like Amazon's Alexa and Apple's Siri, which answer questions upon request.
- **STRONG AI:** These are more complex systems that handle tasks considered human-like. Strong AI is programmed to handle situations where problem-solving is required without human intervention, such as in applications like self-driving cars or hospital operating rooms

1.4 DEEP LEARNING

According to Geoffrey Hinton, widely regarded as the father of deep learning, deep learning (DL) enables computational models composed of multiple processing layers to automatically learn hierarchical representations of data. This approach allows computers to mimic certain aspects of human brain function. Deep learning is a specialized subset of machine learning, which itself is a branch of artificial intelligence (AI).

In deep learning, models are trained to perform tasks such as classification directly from raw data, including images, text, or audio. These models often achieve state-of-the-art performance and, in some cases, can surpass human-level accuracy. Training involves large datasets with labeled examples and neural network architectures composed of many interconnected layers. The system continuously learns and improves by analyzing patterns within the data, enabling it to make increasingly accurate predictions.

Deep learning relies on artificial neural networks, which consist of multiple layers of nodes, or “neurons,” shown in Fig.1.1, that process information and extract meaningful features from complex, unstructured data. It supports various learning paradigms, including supervised, unsupervised, and reinforcement learning. DL underpins many modern AI applications, ranging from chatbots and virtual assistants to self-driving cars and medical image analysis, by enabling systems to automatically identify patterns and make intelligent decisions.

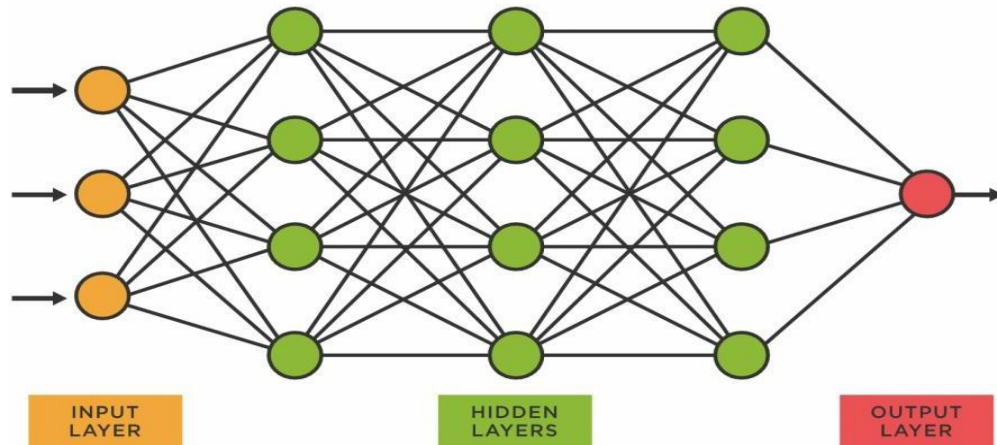


Fig 1.1. Deep learning architecture

1.5 DEEP LEARNING VS MACHINE LEARNING

Machine Learning (ML) and Deep Learning (DL) are important branches of artificial intelligence that are widely applied in data-driven fields, including medical image analysis and disease diagnosis. Machine Learning focuses on enabling computer systems to learn patterns from historical data and use those patterns to make predictions or decisions about new data. Deep Learning, which is a specialized subset of Machine Learning, uses artificial neural networks with multiple layers to automatically learn complex data representations. With the rapid growth of large-scale datasets, these techniques have become essential for extracting meaningful insights and improving predictive accuracy. The fundamental differences between Machine Learning and Deep Learning pipelines are illustrated in Fig. 1.2.

In a traditional Machine Learning pipeline, retinal fundus images first undergo preprocessing steps such as resizing and contrast enhancement. After preprocessing, the system relies on manual feature extraction techniques, including texture analysis, vessel segmentation, and lesion detection, to identify important characteristics such as microaneurysms and hemorrhages. These extracted features are then refined using feature selection methods such as Principal Component Analysis (PCA) before being provided to machine learning classifiers

like Support Vector Machines (SVM), Random Forest, or K-Nearest Neighbors (KNN). Finally, the model produces a classification output that categorizes the severity of diabetic retinopathy into stages such as normal, mild, moderate, severe, or proliferative, as shown in Fig. 1.2.

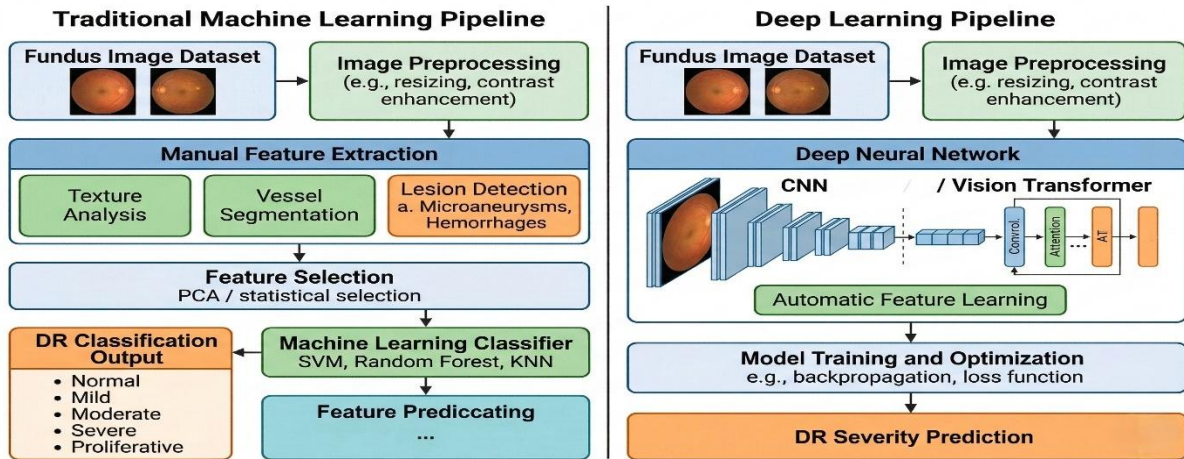


Fig 1.2. Machine learning vs Deep learning

In contrast, Deep Learning follows a more automated and end-to-end learning approach. As illustrated in Fig. 1.2, retinal fundus images are first preprocessed and then directly provided as input to a deep neural network architecture such as a Convolutional Neural Network (CNN) or Vision Transformer (ViT). These models automatically learn hierarchical feature representations from the images without requiring manual feature engineering. During the training phase, the model parameters are optimized using techniques such as backpropagation and loss function minimization. The trained network then predicts the severity level of diabetic retinopathy based on the learned features. This automated feature learning capability allows deep learning models to handle complex visual patterns and large datasets more effectively than traditional machine learning methods.

1.6 NEURAL NETWORKS

Deep learning algorithms aim to reach conclusions in a manner similar to human reasoning by continuously analyzing data within a structured framework. To accomplish this, deep learning employs multi-layered structures of algorithms known as neural networks. These networks are designed based on the architecture of the human brain. Just as humans use their brains to recognize patterns and categorize information, neural networks can be trained to perform similar tasks on data .

Neural networks, also referred to as Artificial Neural Networks (ANNs) or Simulated Neural Networks (SNNs), are a subset of Machine Learning that form the foundation of deep learning techniques. Their structure and naming are inspired by the human brain, mimicking the way biological neurons communicate. ANNs consist of highly parallel systems made up of a large number of interconnected basic processing units. The architecture of a neural network is illustrated in Figure 1.3.

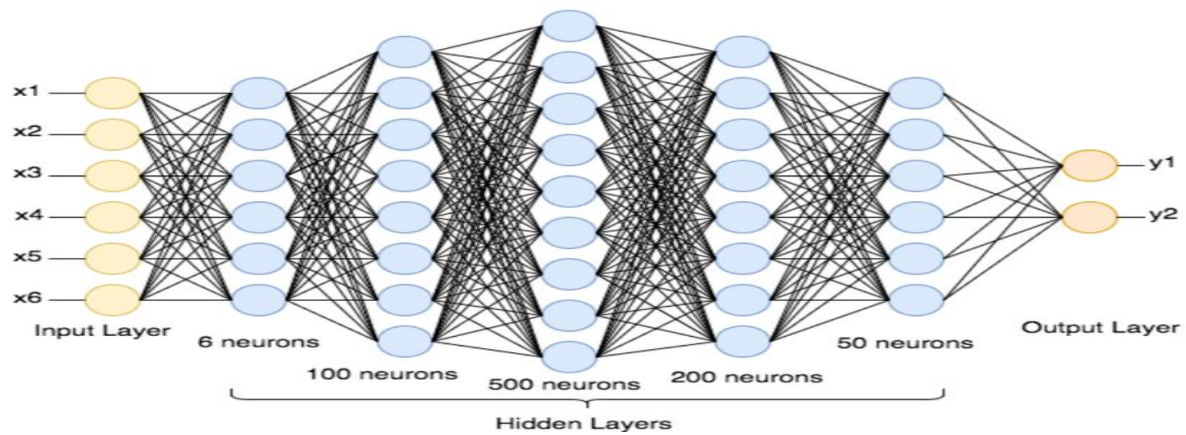


Fig 1.3. Neural network architecture

The layers in a neural network can be viewed as a sequence of filters, moving from coarse to fine, which improves the network's ability to detect patterns and generate accurate outputs. This process mirrors the human brain, which constantly compares new information with previously known objects to make sense of it. Deep neural networks leverage this principle to process complex data effectively.

Neural networks can also be used to cluster or organize unlabeled data based on similarities among the samples. In classification tasks, the network is trained on labeled data, enabling it to categorize the samples into distinct classes with high accuracy.

1.7 IMAGE PROCESSING:

In recent years, deep learning has made a significant impact across various technological fields. One of the most prominent areas is computer vision, which enables computers to interpret and understand images and videos autonomously. Applications such as self-driving cars, biometrics, and facial recognition all rely heavily on computer vision. At the foundation of computer vision lies image processing, which involves representing an image through its dimensions—height and width—based on the total number of pixels. For instance, an image with dimensions 500 x 400 (width x height) contains 200,000 pixels.

A pixel is a point in an image that holds a specific color, shade, or opacity. It is commonly represented in one of the following formats:

- **Grayscale:** A single integer between 0 and 255, where 0 is completely black and 255 is completely white.
- **RGB:** Composed of three integers (0–255) representing the intensity of red, green, and blue.
- **RGBA:** An extension of RGB with an additional alpha channel to indicate opacity.

Image processing involves applying a fixed sequence of operations to each pixel in the image. The processor completes the first set of operations across all pixels before moving on to the next. These operations can be used to extract meaningful information from images, which are typically treated as 2D signals during processing. There are two main approaches: **analogue** and **digital image processing**. Analogue methods are applied to hard copies like photographs and printouts, relying on visual interpretation techniques. Digital image processing, on the other hand, uses computers to manipulate digital images. It generally involves three key phases: pre-processing, enhancement, and display or information extraction.

1.8 VISION TRANSFORMER (ViT)

The Vision Transformer (ViT) is an advanced deep learning architecture that adapts the Transformer model—originally developed for natural language processing (NLP)—to tackle complex computer vision tasks. Unlike conventional Convolutional Neural Networks (CNNs), which rely on convolutional layers to extract local features from images, ViT leverages self-attention mechanisms to model both local and global dependencies across the entire image. This allows the architecture to capture intricate relationships between distant regions of an image, which CNNs may struggle to identify due to their limited receptive fields.

In ViT, an input image is divided into fixed-size, non-overlapping patches. Each patch is flattened and linearly projected into a high-dimensional embedding vector, analogous to word embeddings used in NLP. To retain the spatial arrangement of the patches, a learnable positional embedding is added to each patch embedding. These enriched embeddings are then fed into a standard Transformer encoder, composed of multiple layers of multi-head self-attention and feed-forward neural networks. The self-attention mechanism enables the model to evaluate the relevance of each patch relative to all other patches, effectively capturing long-range interactions and complex structural patterns within the image.

One of the key strengths of ViT is its scalability. By increasing the number of image patches, encoder layers, or attention heads, the model can achieve higher performance, particularly when trained on large-scale datasets. This makes ViT highly flexible and capable of handling a wide variety of image analysis tasks. The architecture has achieved state-of-the-art results in multiple domains, including image classification, object detection, and semantic segmentation, often surpassing CNN-based models in both accuracy and robustness. Furthermore, ViT reduces the inductive biases inherent in convolutional architectures, such as the assumption of spatial locality, allowing the model to learn more generalizable and context-aware features.

Due to its ability to integrate global context with fine-grained local information, the Vision Transformer has become a powerful framework for modern computer vision applications. Its versatility extends to critical fields such as medical imaging, where precise and reliable feature extraction is essential, as well as autonomous driving, surveillance systems, and satellite image analysis. By combining high scalability, global context modeling, and efficient processing, ViT offers a robust and highly effective solution for a wide range of vision-related challenges.

1.9 ARCHITECTURE OF VIT

Vision Transformer (ViT) is a modern deep learning architecture developed for image recognition tasks. Unlike traditional convolution-based models, ViT adapts the Transformer architecture—originally introduced for Natural Language Processing—to process visual data. Conventional image classification approaches such as Convolutional Neural Networks (CNNs) rely on convolutional operations to extract hierarchical spatial features from images. In contrast, Vision Transformers treat an image as a sequence of smaller patches and process them similarly to tokens in a language model, enabling the network to capture both local and global relationships across the entire image, as illustrated in Fig. 1.4.

The image processing pipeline begins by dividing the input retinal fundus image into fixed-size, non-overlapping patches. Each patch is then flattened into a one-dimensional vector and projected into a feature embedding through a linear projection layer. These embeddings collectively form a sequence of patch representations that describe different regions of the image. To preserve spatial information, positional embeddings are added to each patch embedding so that the model can maintain information about the relative positions of patches within the image. Additionally, a special classification token (CLS token) is appended to the

beginning of the patch sequence. During model processing, this token interacts with all patch embeddings and aggregates global contextual information from the entire image. The final representation of the CLS token is then used for image-level classification, as shown in Fig. 1.4.

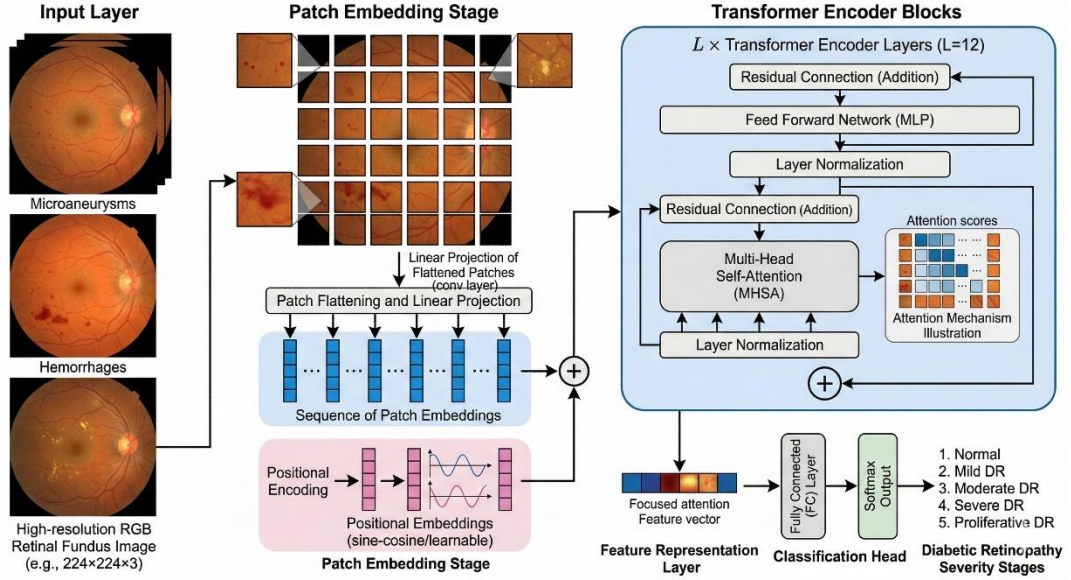


Fig 1.4. Architecture of ViT

The sequence of patch embeddings generated during the patch embedding stage is subsequently processed by the Transformer encoder, which forms the core computational component of the Vision Transformer architecture. The Transformer encoder consists of multiple stacked layers, each containing two principal submodules: the multi-head self-attention mechanism and a position-wise feed-forward neural network. These components work together to progressively refine the representation of the retinal image and capture meaningful relationships among its constituent patches.

The self-attention mechanism is responsible for evaluating the relationships between all image patches simultaneously. Unlike convolutional neural networks, which primarily focus on local neighborhoods through sliding filters, the self-attention mechanism computes interactions between every pair of patches in the input sequence. This allows the model to determine how strongly one region of the retinal image is related to another. In the context of diabetic retinopathy analysis, this capability is extremely valuable because retinal abnormalities such as microaneurysms, hemorrhages, and exudates may appear in spatially distant areas of the retina. The self-attention mechanism allows the model to integrate information from these distributed regions and interpret them collectively when determining disease severity. To further enhance this capability, the Transformer encoder employs a multi-

head self-attention (MHSA) mechanism. Instead of computing a single attention map, the model uses multiple attention heads that operate in parallel. Each attention head learns to focus on different aspects of the retinal image. For example, one attention head may specialize in identifying vascular structures and blood vessel irregularities, while another may focus on detecting lesion clusters or abnormal retinal textures. By combining the outputs of multiple attention heads, the model develops a richer and more comprehensive understanding of the retinal image.

Following the self-attention operation, the resulting feature representations are passed through a feed-forward neural network, typically implemented as a multilayer perceptron (MLP). This component applies nonlinear transformations to the attention outputs, enabling the model to learn more complex and abstract feature representations. The feed-forward network operates independently on each patch embedding, allowing the system to further refine the encoded visual features extracted from the retinal image. Each Transformer encoder layer also incorporates layer normalization and residual connections to improve training stability and efficiency. Layer normalization standardizes the feature distributions within each layer, ensuring that the training process remains stable and preventing issues related to internal covariate shift. Residual connections, on the other hand, create shortcut pathways that allow the model to pass information directly from earlier layers to later layers. This mechanism helps mitigate the vanishing gradient problem and enables the training of deeper transformer architectures without significant degradation in performance.

By stacking multiple Transformer encoder layers, the model gradually learns increasingly complex representations of the retinal image. Early layers may focus on capturing low-level visual patterns such as color variations and texture differences, while deeper layers integrate this information to recognize higher-level structures such as blood vessel networks, lesion distributions, and structural abnormalities within the retina.

CHAPTER II

LITERATURE SURVEY

2.1 INTRODUCTION TO THE SURVEY

A survey introduction establishes the context, scope, and purpose of reviewing existing research, emphasizing the significance of the topic and identifying gaps that require further investigation. In the domain of diabetic retinopathy (DR), surveys are essential for summarizing advancements in automated diagnostic methods, particularly those employing artificial intelligence (AI) and deep learning techniques. DR is a microvascular complication of diabetes mellitus and a leading cause of vision impairment and blindness worldwide. Early detection and accurate grading of DR are crucial for preventing irreversible vision loss and enabling timely clinical interventions.

Traditional manual screening of retinal fundus images is time-consuming, subjective, and prone to variability, which has motivated the development of automated, intelligent solutions. Over recent years, numerous studies have proposed deep learning-based approaches, including convolutional neural networks (CNNs), transfer learning frameworks, and hybrid architectures, aimed at improving the accuracy and efficiency of DR detection and classification. Surveys provide a comprehensive overview of these methods by analyzing model architectures, datasets, evaluation metrics, and application domains. They help identify the strengths, limitations, and emerging trends in the field, as well as challenges such as limited annotated data, interpretability of black-box models, generalization across diverse populations, and integration into real-world clinical workflows.

By systematically reviewing and comparing existing literature, surveys guide the development of more robust, accurate, and clinically applicable models. They also facilitate knowledge transfer across interdisciplinary domains, including computer vision, ophthalmology, and medical imaging, ultimately contributing to enhanced patient outcomes. This survey specifically aims to provide a detailed analysis of recent advancements in AI-based DR diagnosis, offering insights into state-of-the-art techniques, datasets, challenges, and future research directions for clinical implementation. Furthermore, it highlights the growing importance of explainable AI in enhancing transparency and clinician trust in automated systems. It also emphasizes the need for standardized evaluation protocols to ensure fair comparison of models across different datasets. Ultimately, the survey supports the development of scalable, reliable, and widely adoptable DR diagnostic solutions in healthcare systems.

2.2 LITERATURE SURVEY

2.2.1 Developing a Transparent Diagnosis Model for Diabetic Retinopathy Using Explainable AI by Tariq Shahzad.et.al.

Diabetic retinopathy (DR) is a leading cause of vision loss and poses significant challenges for early detection due to its varied and subtle symptoms. Diagnosis traditionally depends on expert analysis of retinal fundus images, which can be time-consuming, inconsistent, and reliant on specialist availability. Although conventional AI methods have improved detection accuracy, their lack of transparency limits clinical trust and usability. Explainable AI (XAI) overcomes this issue by offering interpretable insights into model decisions, enabling better understanding and confidence among healthcare professionals. The proposed XAI-based model enhances diagnostic performance while providing clear reasoning for its predictions. Experimental evaluation, including convergence analysis, shows that the model achieves up to 94% higher accuracy compared to traditional approaches. Additionally, it reduces loss and improves reliability, making it more suitable for real-world clinical implementation. Furthermore, the model facilitates better clinical decision-making by highlighting important regions in retinal images associated with disease progression. This interpretability supports early intervention and promotes wider adoption of AI-based diagnostic tools in healthcare systems.

2.2.2 Advancing diabetic retinopathy diagnosis with fundus imaging: A comprehensive survey of computer-aided detection, grading and classification methods by S. Prathibha.et.al

The rising global incidence of Diabetic Retinopathy (DR) emphasizes the importance of developing reliable computer-aided diagnosis (CAD) systems for early detection and accurate assessment. DR, a complication of diabetes, can lead to severe vision impairment if not identified promptly, with key indicators such as microaneurysms, hemorrhages, and exudates visible in retinal images. However, analyzing fundus images is challenging due to complex patterns, variations in lighting, and patient-specific differences. Traditional CAD systems rely on handcrafted features and classical machine learning, which often struggle to generalize across diverse datasets. Many existing systems also focus on detecting only specific lesions, limiting their effectiveness in real clinical scenarios where multiple abnormalities may coexist. This narrow scope increases diagnostic time and workload. Recent advancements in deep learning address these issues by automatically learning complex features from images. As a result, modern models can detect multiple abnormalities more accurately and improve overall diagnostic performance.

2.2.3. Application of AI for Classification, Segmentation, Early Detection, Early Diagnosis, and Grading of DR by G. Rajarajeshwari.et.al

Diabetic Retinopathy (DR) is a major cause of vision impairment worldwide, making early and accurate diagnosis essential for effective treatment. Artificial Intelligence (AI) has emerged as a powerful tool for improving DR detection, classification, and risk assessment. This review analyzes AI-based approaches across key areas such as classification, detection, early diagnosis, segmentation, and severity grading. It presents a structured workflow and highlights real-world case studies demonstrating the practical application of AI in clinical settings. The study also emphasizes the importance of Explainable AI (XAI) in enhancing model transparency, validation, and generalization. Additionally, it discusses commonly used datasets, evaluation metrics, and existing research challenges. Key limitations include data variability, preprocessing issues, and challenges in integrating multiple data sources. Addressing these concerns can lead to more accurate, reliable, and clinically applicable AI systems for DR management.

2.2.4. Deep Learning in Automatic DR Detection and Grading Systems by Israa Y. Abushawish et.al

Diabetic Retinopathy (DR) is a major cause of vision loss worldwide, especially among diabetic patients, highlighting the need for efficient automated detection systems. Traditional machine learning methods are limited due to their dependence on handcrafted features and inability to handle complex retinal patterns. Deep learning, particularly convolutional neural networks (CNNs), has significantly improved DR detection by automatically extracting hierarchical features from fundus images. This study reviews the evolution from conventional techniques to advanced models using transfer learning, end-to-end learning, and hybrid approaches. These methods address challenges such as limited data, image variability, and the need for generalization across populations. The research evaluates 26 pre-trained CNN models to compare their effectiveness in detection and severity grading. It also emphasizes the importance of interpretability in clinical applications. Grad-CAM visualization is used to highlight important image regions and explain model decisions. This enhances trust and bridges the gap between AI predictions and clinical understanding. Overall, combining deep learning with explainable AI improves accuracy, reliability, and real-world applicability in DR diagnosis.

2.2.5. Deep Diabetic: An Identification System of Diabetic Eye Diseases by Arwa Albelaihi.et.al

Deep Learning (DL) has shown strong potential in medical imaging for tasks such as diagnosis, detection, and classification of diseases. This study proposes a multi-class deep learning framework, DeepDiabetic, to identify four diabetic eye diseases: Diabetic Retinopathy (DR), Diabetic Macular Edema (DME), glaucoma, and cataract. The model was trained and evaluated on 1228 images collected from six public datasets, with data augmentation techniques applied to improve performance. Five deep learning architectures were analyzed, including EfficientNetB0, VGG16, and ResNet-based models with GRU variations. A comparative analysis was conducted to evaluate their effectiveness on multi-class fundus images. Results showed that EfficientNetB0 outperformed all other models in terms of accuracy, precision, recall, and AUC. It also achieved better performance than earlier approaches like Fast-RCNN and InceptionResNet. The study highlights the effectiveness of the DeepDiabetic framework for accurate disease classification. Overall, it provides a reliable and efficient solution for early detection and clinical management of diabetic eye diseases.

2.2.6. Diabetic Retinopathy Classification With Deep Learning via Fundus Images: A Short Survey by Shanshan Zhu et al.

Diabetic retinopathy (DR) is a major complication of diabetes and a leading cause of vision loss worldwide, making early detection and accurate grading essential. Traditional diagnosis relies on manual examination of fundus images, which is time-consuming and prone to variability. Deep learning (DL)-based systems have emerged as effective solutions, offering improved accuracy, efficiency, and scalability by automatically learning complex image features. This review analyzes 94 research studies published between 2018 and 2023 to identify trends and challenges in DL-based DR detection. It highlights the importance of transfer learning in handling limited and imbalanced datasets. Popular CNN architectures such as ResNet and VGGNet are widely used, along with datasets like APTOS 2019 and EyePACS for benchmarking. Lightweight models like MobileNet and SqueezeNet are also explored for resource-constrained environments. Despite achieving high performance, challenges such as interpretability, trust, and cost-effective deployment remain. Addressing these issues is crucial for integrating DL systems into real-world clinical practice. Furthermore, improving model explainability and standardizing evaluation protocols can enhance clinician confidence and adoption. Future research should also focus on developing robust, scalable, and affordable solutions for widespread healthcare implementation.

2.2.7. Automated Diagnosis of Diabetic Retinopathy Using Deep Learning: On the Search of Segmented Retinal Blood Vessel Images for Better Performance by Mohammad B Khan.et.al

Diabetic retinopathy (DR) is a serious retinal disease and a leading cause of blindness, requiring timely and accurate diagnosis to prevent vision loss. Traditional manual screening methods are often limited by subjectivity and human error, leading to possible misdiagnosis. Deep learning-based automated systems have emerged as effective solutions for early detection and treatment. This study compares two architectures, Inception v3 and DenseNet-121, using both original fundus images and segmented blood vessel images. Results show that both models achieve high accuracy (≥ 0.8) when trained on original-colored images. However, performance drops significantly (just above 0.6) when using only segmented vessel images. This indicates that original fundus images provide richer and more informative features for DR detection. The findings suggest that relying solely on segmented images is less effective. Therefore, future systems should prioritize complete retinal images for improved diagnostic accuracy.

2.2.8. Enhancing Early Detection of Diabetic Retinopathy Through the Integration of Deep Learning Models and Explainable Artificial Intelligence

The rising prevalence of Diabetic Retinopathy (DR) highlights the need for reliable automated systems for early detection and prevention of vision loss. DR is a retinal disorder caused by diabetes that can lead to blindness if not diagnosed early, especially since initial symptoms are often unnoticed. Deep learning techniques, particularly convolutional neural networks (CNNs), have enabled effective analysis of retinal fundus images by extracting complex features. Various transfer learning models such as DenseNet121, Xception, ResNet50, VGG16, VGG19, and InceptionV3 have been used to improve classification accuracy. Traditional methods like Support Vector Machine (SVM) and Recurrent Neural Networks (RNN) were also evaluated. Multi-label classification using Softmax and categorical cross-entropy was applied to enhance prediction reliability across DR severity levels. Experimental results demonstrated strong performance in terms of accuracy, precision, and recall. Among all models, Xception achieved the highest accuracy of around 82%, making it the most effective in this study. Furthermore, the use of transfer learning reduces training time and improves model generalization on limited datasets. These advancements contribute to developing efficient, scalable, and clinically applicable DR detection systems.

2.2.9. Prediction and analysis of risk factors for diabetic retinopathy based on machine learning and interpretable models by Wang et al.

An interpretable machine learning framework was developed to predict the risk of Diabetic Retinopathy (DR) using 1,000 early-warning samples from a clinical dataset. Data preprocessing included handling missing values, removing outliers, and selecting relevant features using the information gain method. Multiple models such as SVM, Random Forest, XGBoost, GBT, and CatBoost were evaluated for performance. Among them, CatBoost achieved the best results with 82.5% accuracy and 89% AUC, indicating strong predictive capability. To enhance interpretability, SHAP analysis was used to explain feature importance and model decisions. Key risk factors identified included renal dysfunction, high HbA1c levels, and nephropathy-related indicators. Some blood and liver function markers showed negative correlations, suggesting potential protective roles. Additional factors like BMI, blood pressure, and fatty liver also influenced DR risk. Overall, the study highlights the importance of interpretable models in improving clinical trust and decision-making.

2.2.10. Convolutional Neural Network (CNN) approach for diagnosing and classifying the severity of diabetic retinopathy by Pratt et al.

A Convolutional Neural Network (CNN)-based framework was developed for classifying Diabetic Retinopathy (DR) using over 80,000 retinal fundus images from the Kaggle EyePACS dataset. The dataset included five DR severity levels: No DR, Mild, Moderate, Severe, and Proliferative DR. Preprocessing steps such as color normalization and resizing images to 512×512 pixels were applied to ensure consistency. Data augmentation techniques like rotation, flipping, and zooming were used to improve generalization and reduce overfitting. The CNN architecture consisted of multiple deep convolutional layers to capture complex visual and pathological features. Regularization methods such as dropout and L2 regularization were incorporated to enhance model robustness. The leaky ReLU activation function was used to maintain gradient flow and improve learning. A class-balancing strategy was applied to handle dataset imbalance across DR severity levels. This ensured better representation of minority classes during training. Overall, the framework improves classification performance and reliability in DR detection. Furthermore, the use of a large-scale dataset enhances the model's ability to generalize across diverse patient conditions. This approach supports the development of accurate and scalable automated screening systems for clinical applications.

2.2.11. A Privacy-Preserving Approach Using Deep Learning Models for Diabetic Retinopathy Diagnosis

Privacy-preserving deep learning techniques have been explored for automated diabetic retinopathy detection using retinal fundus images. Convolutional neural network architectures such as ResNet50 and DenseNet169 were applied to extract disease-related features from retinal datasets. Integration of data protection mechanisms, including encryption and differential privacy, allowed the system to protect sensitive medical information while maintaining diagnostic capability. Experimental observations indicated that privacy-preserving strategies can be incorporated into deep learning models without significantly affecting classification performance. However, reliance on convolutional architectures mainly emphasizes local feature extraction and may limit the model's ability to capture long-range spatial relationships within retinal structures. In addition, focus was primarily placed on prediction accuracy rather than comprehensive multi-stage classification of diabetic retinopathy severity. Vision Transformer architecture proposed in this work addresses these limitations by dividing retinal images into patches and applying self-attention mechanisms to capture global contextual relationships. Such capability improves feature representation and enables effective multi-class classification of diabetic retinopathy stages.

2.2.12. A Systematic Review on Fundus Image-Based Diabetic Retinopathy Detection and Grading: Current Status and Future Directions

Comprehensive investigation of machine learning and deep learning techniques has been conducted for automated diabetic retinopathy detection and grading using retinal fundus images. Various components of the detection pipeline were analyzed, including publicly available datasets, preprocessing methods, feature extraction strategies, classification algorithms, and evaluation metrics. Findings indicate that many existing systems depend heavily on convolutional neural networks and conventional machine learning techniques for retinal image analysis. Despite improvements in diagnostic accuracy compared with manual screening, several challenges remain unresolved. Issues such as dataset imbalance, variability in retinal image quality, limited model generalization, and reduced interpretability continue to affect automated diagnostic systems. Convolution-based architectures also tend to emphasize localized spatial features while overlooking global contextual relationships within retinal images. Vision Transformer-based framework proposed in this work addresses these limitations by processing retinal images as sequences of patches and learning contextual dependencies through self-attention mechanisms.

2.2.13. An Insight on the Timely Diagnosis of Diabetic Retinopathy Using Traditional and AI-Driven Approaches

Comparison between traditional diagnostic methods and artificial intelligence-based approaches has been investigated for diabetic retinopathy detection. Manual examination of retinal fundus images by ophthalmologists remains a common diagnostic approach but often requires significant time and may introduce human interpretation errors. Machine learning models introduced partial automation but required handcrafted feature extraction, which increased computational complexity and limited scalability. Deep learning techniques improved automated feature learning and enhanced diagnostic performance, yet many models still function as black-box systems with limited interpretability in clinical environments. Increasing interest in explainable artificial intelligence highlights the importance of transparent and reliable diagnostic systems for healthcare applications. However, many existing approaches remain dependent on convolutional architectures that primarily capture local features and may fail to represent broader contextual relationships across retinal images. Vision Transformer architecture utilized in the proposed system overcomes these challenges by employing self-attention mechanisms capable of modeling long-range dependencies across image regions.

2.2.14. Enhancing Early-Stage Diabetic Retinopathy Detection Using a Weighted Ensemble of Deep Neural Networks

Ensemble-based deep learning strategies have been developed to improve early detection of diabetic retinopathy using retinal fundus images. Multiple convolutional neural network models, including Inception-V3, VGG16, and a custom CNN architecture, were combined to enhance classification performance. Weighted averaging techniques were applied to integrate predictions from different models, resulting in improved accuracy, precision, recall, and F1-score compared with individual networks. Ensemble methods demonstrate strong performance in retinal image classification tasks; however, integration of multiple neural networks significantly increases computational complexity and training cost. Requirement of several deep models also reduces efficiency and scalability in practical medical systems. Convolutional architectures used within ensemble frameworks mainly focus on local feature extraction and may struggle to capture global relationships among retinal structures. Vision Transformer-based model proposed in this research addresses these issues by modeling long-range dependencies using self-attention mechanisms.

2.2.15. Optical imaging for diabetic retinopathy diagnosis and detection using ensemble models

Ensemble deep learning approaches have also been applied for diabetic retinopathy detection using retinal imaging techniques. Feature extraction from retinal fundus images was performed using multiple convolutional neural network architectures, including VGG19, ResNet50, and InceptionV3. Combination of these models allowed improved identification of retinal abnormalities such as microaneurysms, hemorrhages, and neovascularization. Ensemble learning helps integrate complementary strengths of different neural network architectures and can improve overall classification accuracy. Despite improved performance, ensemble frameworks require extensive computational resources and complex training processes due to the use of multiple deep models. CNN-based architectures primarily focus on localized spatial patterns and may not fully capture global contextual dependencies present within retinal images. Limitations of such approaches highlight the need for advanced architectures capable of learning global feature relationships more efficiently. Vision Transformer architecture proposed in this work processes retinal images as sequences of patches and utilizes self-attention mechanisms to capture both local and global information. Resulting model provides improved feature representation and enables reliable multi-class classification of diabetic retinopathy severity levels.

Table 2.1 Literature Survey Table

Ref No	Title	Authors	Methods Used	Advantages	Disadvantages
[1]	Developing a Transparent Diagnosis Model for Diabetic Retinopathy Using Explainable AI	Shahzad et al.	CNN + LIME (XAI)	Improves interpretability, strong diagnostic accuracy	Computationally expensive, small dataset
[2]	Diabetic Retinopathy Classification with Deep Learning via Fundus Images: A Short Survey	Zhu et al.	CNN, ResNet, Inception, DenseNet	Provides concise review of CNN-based approaches	Lacks detailed experiments, limited coverage
[3]	Optical Imaging for Diabetic Retinopathy Diagnosis and Detection Using Ensemble Models	Pavithra et al.	Optical imaging + Ensemble ML	Non-invasive, enhances accuracy through ensembles	Requires costly imaging equipment
[4]	Enhancing Early Detection of DR through DL Models and Explainable AI	Alavee et al.	CNN + Ensemble + Grad-CAM, LIME	High accuracy, interpretable outputs	Complex architecture, longer training time
[5]	Diagnosis of Diabetic Retinopathy Using Deep Neural Networks	Gao et al.	InceptionV3, Inception@4	Clinical reliability, lesion-focused detection	Lower accuracy (88.7%) than modern methods
[6]	DeepDiabetic: An Identification System of Diabetic Eye Diseases Using DNN	Albelaihi et al.	EfficientNetB0, VGG16, ResNet152V2, GRU hybrids	Multi-disease detection, very high accuracy (0.9876)	Dataset small (1228 images), limited diversity
[7]	Deep Learning in Automatic DR Detection and Grading Systems: Survey and Comparison	Abushawish et al.	CNN, ResNet, U-Net, GANs	Comparative analysis of existing DR models	Overfitting risks, interpretability challenges
[8]	Enhancing Early-Stage DR Detection	Nazir et al.	Inception-v3, VGG16, Custom CNN	High performance (95%)	Training is complex, high

	Using a Weighted Ensemble of DNNs		(weighted ensemble)	accuracy), strong early detection	computational demand
[9]	Prediction and Analysis of Risk Factors for DR	Wang et al.	CatBoost ML + SHAP	Identifies and explains DR risk factors	Not image-based, limited clinical deployment
[10]	Convolutional Neural Networks for Diabetic Retinopathy	Pratt et al.	CNN (custom, Kaggle dataset)	First 5-class CNN study, high sensitivity (95%)	Overall accuracy 75%, weak on severe cases
[11]	Application of AI for Classification, Segmentation, Detection, and Grading of DR: A Review	Rajarajeshwari et al.	CNN, ML (SVM, RF), Deep ensemble models	Comprehensive survey of AI in DR	Dataset imbalance, long training times
[12]	An Insight on the Timely Diagnosis of DR Using Traditional and AI-Driven Approaches	Asif et al.	Fundus photography, OCT, CNN, ML	Reviews both manual and AI-driven diagnosis	Manual is subjective, OCT is costly
[13]	Advancing Diabetic Retinopathy Diagnosis with Fundus Imaging: A Comprehensive Survey	Prathibha et al.	Fundus imaging + DL-based methods	Broad survey, highlights advancements	Dependent on image quality, limited scalability
[14]	A Systematic Review on Fundus Image-Based DR Detection and Grading	Ikram et al.	CNN, U-Net, ML classifiers	Covers multiple approaches, identifies gaps	Only a review, no experimental validation
[15]	A Privacy-Preserving Approach Using DL Models for DR Diagnosis	Ng et al.	CNN + Federated Learning + Encryption	Protects patient data privacy	Slight accuracy reduction vs central training

2.3 EXISTING SOLUTIONS

The current system for diabetic retinopathy (DR) classification primarily employs Convolutional Neural Networks (CNNs) to analyze retinal fundus images and identify different stages of the disease. CNNs are highly effective at capturing local spatial features such as microaneurysms, hemorrhages, and exudates, which are critical for accurate DR detection. To enhance transparency and build clinical trust, the system incorporates Explainable AI (XAI) techniques, notably Grad-CAM, which produces visual heatmaps that highlight the regions of an image most influential to the model's predictions.

This approach has demonstrated promising results, achieving a diagnostic accuracy of up to 94% and outperforming several traditional and deep learning-based methods, bridging the gap between automated analysis and clinical applicability.

Despite these advantages, the existing system has several notable limitations. CNNs primarily focus on local features and often fail to capture long-range dependencies or global structural patterns within complex retinal images, which are important for accurate staging of DR. While Grad-CAM aids interpretability, it is a post hoc tool and does not fully expose the internal reasoning of the model.

Additionally, CNNs are sensitive to variations in image quality, noise, and acquisition conditions, which can hinder their generalization across diverse datasets. The deep CNN architectures also require substantial computational resources and careful hyperparameter tuning, yet they largely remain black-box models. Training on limited datasets increases the risk of overfitting and reduces scalability, particularly when processing high-resolution images. These challenges underscore the need for advanced models capable of capturing both local and global image features, providing improved interpretability, and ensuring robust performance across varied clinical scenarios.

2.4. LIMITATIONS IN THE EXISTING SYSTEM

1. Local Feature Focus

Convolutional Neural Networks (CNNs) are widely used in medical image analysis because of their strong ability to detect local patterns and small visual features. In the context of diabetic retinopathy detection, CNNs can successfully identify minute abnormalities such as microaneurysms, hemorrhages, and exudates by applying convolutional filters over small receptive fields. However, this architecture inherently focuses on localized regions of an image and may struggle to capture broader spatial relationships between distant areas of the retina.

As a result, CNN-based models may fail to fully understand the global structure of retinal images, including complex vascular patterns and long-range dependencies between lesions. This limitation can reduce the model's ability to accurately represent the overall disease progression and severity in some cases.

2. Interpretability Constraints

Although interpretability methods such as Gradient-weighted Class Activation Mapping (Grad-CAM) are often applied to CNN models to highlight regions that influence predictions, these explanations are considered post hoc in nature. This means that the interpretability mechanism is applied after the model has already made its prediction rather than being inherently integrated into the model's architecture. Consequently, these visual explanations only provide partial insight into the model's reasoning process and may not fully reflect how internal feature representations contribute to the final decision. In clinical environments, where transparency and reliability are critical, this lack of intrinsic interpretability can reduce the level of trust healthcare professionals place in automated diagnostic systems.

3. Generalization Issues

CNN-based diagnostic systems are often trained on datasets with similar imaging conditions, limiting their adaptability. Retinal images from different sources vary in resolution, lighting, contrast, and noise due to differences in devices and acquisition methods. As a result, models tend to learn dataset-specific patterns rather than general features. This causes performance degradation when applied to images with different visual characteristics. For instance, models trained on high-quality images may fail on low-quality or poorly lit inputs. Such lack of generalization restricts their reliability in real-world clinical environments.

4. Model Complexity

Modern CNN architectures for medical image analysis contain multiple layers and millions of parameters, enabling extraction of complex features from retinal images. However, this also leads to high computational complexity and resource requirements. Training such models demands powerful hardware like GPUs and large volumes of labeled data. In medical imaging, obtaining annotated datasets is difficult due to the need for expert ophthalmologists. Additionally, extensive hyperparameter tuning increases development time and cost. Processing high-resolution images further adds to memory usage and computational overhead during training and inference.

CHAPTER III

PROPOSED SYSTEM

3.1 PROBLEM STATEMENT

Diabetic retinopathy (DR) has emerged as one of the most significant causes of vision impairment and blindness among adults worldwide, largely driven by the rapid increase in diabetes prevalence across both developed and developing nations. Diabetes mellitus affects the body's ability to regulate blood glucose levels, and prolonged hyperglycemia can damage small blood vessels throughout the body, including those in the retina. Over time, this damage leads to progressive retinal abnormalities such as microaneurysms, hemorrhages, exudates, and abnormal blood vessel growth, which characterize the different stages of diabetic retinopathy. According to global health reports from the World Health Organization and the International Diabetes Federation, hundreds of millions of people are currently living with diabetes, and a significant proportion of these individuals are at risk of developing diabetic retinopathy during their lifetime. As the global diabetic population continues to grow, the burden of DR-related visual impairment is expected to increase substantially.

Early detection and timely medical intervention are critical for preventing severe vision loss caused by diabetic retinopathy. In its early stages, DR may not produce noticeable symptoms, making regular retinal screening essential for patients with diabetes. If the disease is identified early, treatments such as laser therapy, intravitreal injections, or surgical procedures can effectively slow disease progression and preserve vision. However, the effectiveness of these interventions depends heavily on the availability of reliable screening systems capable of identifying retinal abnormalities before irreversible damage occurs.

Currently, most diabetic retinopathy screening programs rely on manual examination of retinal fundus images by trained ophthalmologists or retinal specialists. During this process, clinicians analyze fundus photographs to detect pathological features and determine the severity stage of the disease. While this manual diagnostic approach is considered the clinical gold standard, it faces several limitations. The growing number of patients requiring screening places a significant burden on healthcare systems, particularly in regions with limited medical resources. Manual analysis of retinal images is time-consuming and requires extensive clinical expertise, which makes large-scale screening difficult to implement efficiently. In many rural or underserved areas, the shortage of ophthalmologists results in delayed diagnoses or inadequate screening coverage. Additionally, manual interpretation of retinal images may be subject to variability among clinicians, leading to inconsistencies in diagnosis. Differences in

experience, fatigue, or interpretation criteria can contribute to inter-observer variability, increasing the risk of both under-diagnosis and over-diagnosis of diabetic retinopathy.

To address these challenges and improve the scalability of screening programs, researchers have increasingly explored the use of computer-aided diagnostic systems powered by artificial intelligence and deep learning. AI-based systems have demonstrated significant potential in automating medical image analysis tasks by learning complex patterns from large datasets. In the field of ophthalmology, deep learning techniques have been widely applied to retinal image classification, enabling automated detection of diabetic retinopathy with promising accuracy levels. These systems can process large volumes of retinal images rapidly and consistently, assisting clinicians by providing preliminary diagnostic assessments and reducing the workload associated with manual screening.

Among the various deep learning approaches, Convolutional Neural Networks (CNNs) have become the foundation of many automated DR detection systems. CNNs are particularly effective in extracting hierarchical spatial features from images through layers of convolution, pooling, and nonlinear activation functions. In retinal image analysis, CNN models can detect subtle local features such as microaneurysms, hemorrhages, and lipid deposits that indicate early stages of diabetic retinopathy. By learning these features directly from labeled training data, CNN-based models can achieve high levels of classification accuracy and have been widely adopted in research and commercial medical imaging solutions.

Despite these advancements, CNN-based diabetic retinopathy detection systems still face several significant challenges that limit their effectiveness in real-world clinical applications. One of the primary limitations of CNN architectures is their strong reliance on local feature extraction. Convolutional filters analyze small neighborhoods of pixels at a time, which allows the model to detect fine-grained details but often restricts its ability to capture long-range dependencies between distant regions of the image. In retinal images, disease indicators may appear in multiple spatially separated locations, and understanding the relationship between these regions can be crucial for determining disease severity. CNNs may struggle to effectively capture such global contextual relationships because their receptive fields expand gradually across layers.

Another major challenge associated with CNN-based systems is their sensitivity to variations in imaging conditions. Retinal images collected from different hospitals or screening programs often vary in resolution, illumination, contrast, and noise levels. CNN models trained on a specific dataset may learn patterns that are strongly dependent on those particular imaging conditions. As a result, their performance may decline when applied to images captured using

different devices or under different clinical settings. This lack of robustness can reduce the reliability of CNN-based systems in real-world deployment scenarios.

Furthermore, CNN models typically require large amounts of high-quality labeled training data to achieve optimal performance. Obtaining such datasets in the medical domain can be challenging because annotation of retinal images requires expert knowledge and considerable time investment from trained ophthalmologists. Limited availability of labeled data can lead to overfitting, where the model performs well on training data but fails to generalize to new unseen images. This issue becomes particularly problematic when developing systems intended for large-scale clinical deployment across diverse patient populations.

Another limitation of CNN architectures is their computational complexity. Deep CNN models often contain numerous convolutional layers and millions of parameters, requiring substantial computational resources for both training and inference. Processing high-resolution retinal images with deep CNN architectures can be computationally expensive and time-consuming. These requirements may pose challenges for deploying CNN-based diagnostic systems in healthcare environments with limited computational infrastructure.

Additionally, CNN models are frequently criticized for their lack of interpretability. In medical applications, clinicians must be able to understand how an AI system arrives at its predictions in order to trust and effectively use it in clinical decision-making. However, CNN models often function as black-box systems, providing predictions without clearly explaining the reasoning behind them. Although techniques such as Grad-CAM can provide visual explanations, these methods are generally applied after the prediction and do not fully reveal the internal decision-making process of the model.

Considering these limitations, there is a growing need for more advanced deep learning architectures that can overcome the shortcomings of traditional CNN-based systems. An ideal automated DR detection model should be capable of capturing both local retinal abnormalities and global contextual relationships within retinal images. It should also provide interpretable predictions that allow clinicians to understand the reasoning behind diagnostic decisions. Furthermore, the system should demonstrate strong generalization ability across diverse datasets and maintain computational efficiency suitable for large-scale screening programs.

Therefore, modern architectures such as Vision Transformers have emerged as promising alternatives for medical image analysis tasks. By leveraging attention mechanisms that analyze relationships between all regions of an image simultaneously, transformer-based models can capture complex spatial dependencies more effectively than traditional

convolutional networks. Integrating such architectures into diabetic retinopathy detection systems offers the potential to improve diagnostic accuracy, enhance interpretability, and support scalable automated screening solutions capable of addressing the growing global burden of diabetic eye disease.

3.2 PROPOSED WORK

The proposed work introduces an advanced deep learning framework for automated diabetic retinopathy (DR) diagnosis using the Vision Transformer (ViT) architecture. The system aims to improve both the accuracy and interpretability of retinal disease detection by replacing traditional convolutional neural network (CNN) approaches with transformer-based learning mechanisms. Diabetic retinopathy is one of the leading causes of vision impairment among diabetic patients, and early diagnosis plays a crucial role in preventing permanent blindness. However, manual screening of retinal fundus images by ophthalmologists can be time-consuming and requires significant expertise. Automated systems can assist healthcare professionals by providing faster and consistent preliminary diagnoses.

The primary motivation behind the proposed framework is to address the limitations associated with conventional CNN-based diagnostic models. Although CNNs have achieved considerable success in medical image analysis, they primarily focus on extracting local features through convolutional filters. This localized feature extraction often limits the ability of CNN models to understand global contextual relationships within retinal images. In contrast, the Vision Transformer architecture utilizes a self-attention mechanism that allows the model to analyze relationships between all regions of an image simultaneously. This capability enables the system to capture both local lesion details and global structural patterns within the retina, which are essential for accurate diabetic retinopathy classification.

In addition to improving detection accuracy, the proposed system also emphasizes explainability and transparency. Medical diagnostic systems must provide interpretable outputs to gain the trust of healthcare professionals. Therefore, the framework incorporates both inherent attention visualization and post-hoc explanation techniques to provide insights into how the model arrives at its predictions. By combining transformer-based learning with explainable artificial intelligence methods, the system aims to deliver a reliable and clinically supportive DR diagnosis tool.

1.Motivation for Using Vision Transformers

Traditional deep learning models for medical image classification have primarily relied on convolutional neural networks. CNN architectures have demonstrated strong performance in recognizing visual patterns such as edges, textures, and localized anomalies. In diabetic retinopathy detection, CNN models are capable of identifying small retinal abnormalities such as microaneurysms, hemorrhages, hard exudates, and cotton wool spots. These abnormalities often serve as early indicators of diabetic retinal damage.

Despite these advantages, CNN-based models possess certain inherent limitations. One of the key limitations is their restricted receptive field, which causes them to primarily focus on local image regions. Although deeper CNN layers can capture broader patterns, the model still relies heavily on sequential convolution operations that may not effectively capture long-range relationships between distant regions of the retina. For instance, certain DR indicators may appear in multiple scattered regions of the retinal image, and understanding their spatial relationships can be important for determining disease severity.

The Vision Transformer architecture offers a promising alternative to overcome these challenges. Inspired by transformer models used in natural language processing, ViT processes images as sequences of smaller image patches. Instead of relying on convolutional filters, the model employs a self-attention mechanism that evaluates the relationships between all patches simultaneously. This allows the model to identify complex spatial dependencies and contextual interactions across the entire image. For example, abnormalities located near the optic disc and those appearing in peripheral retinal regions may collectively indicate a more advanced stage of diabetic retinopathy. A transformer-based model can effectively capture such spatial relationships because it evaluates all patches within a global attention framework. This global perspective enables the model to build a more holistic understanding of retinal structures and disease patterns.

Another advantage of Vision Transformers is their ability to produce attention maps that highlight important image regions contributing to the final prediction. These attention maps can be visualized to help clinicians understand which retinal areas influenced the diagnostic decision. This level of transparency is particularly important in medical applications where explainability and trust are critical. Furthermore, Vision Transformers have demonstrated strong performance when trained on large datasets and have shown better scalability compared to traditional CNN models. As more retinal image datasets become available, transformer-based models can leverage this data to learn richer and more generalized feature representations.

2. System Architecture Overview

Retinal fundus images used in the proposed system are obtained from well-established and publicly available medical imaging datasets such as APTOS, EyePACS, and Messidor. These datasets are widely recognized in ophthalmology research and contain thousands of high-resolution retinal images captured using specialized fundus cameras during clinical examinations. The datasets collectively represent the full spectrum of diabetic retinopathy (DR) progression, including images of healthy retinas as well as those exhibiting mild, moderate, severe, and proliferative stages of the disease. Each image in these datasets is carefully annotated by expert ophthalmologists based on the severity of retinal abnormalities present. These annotations play a crucial role in supervised learning, as they allow the deep learning model to learn the relationship between visual retinal features and their corresponding disease stages. The diversity of images in these datasets, including variations in lighting conditions, patient demographics, camera types, and image resolutions, helps the model learn robust features and improves its ability to generalize across different clinical environments.

Before the images are used for model training, a comprehensive preprocessing stage is applied to ensure that the data is standardized and suitable for the Vision Transformer architecture. Retinal images collected from different sources often vary in size, brightness, and contrast, which can negatively affect the learning process if not normalized. Therefore, the preprocessing phase begins by resizing all images to a consistent resolution that matches the input requirements of the Vision Transformer model. Normalization techniques are applied to scale pixel intensity values to a uniform range, which stabilizes the training process and improves convergence. Additionally, contrast enhancement methods are used to emphasize important retinal structures such as blood vessels, lesions, and abnormal tissue regions. Noise reduction techniques are also implemented to remove unwanted artifacts that may appear due to camera limitations or imaging conditions. These preprocessing operations collectively ensure that the input images maintain consistent visual quality and highlight medically relevant features for accurate analysis.

Once the images are preprocessed, they are prepared for processing by the Vision Transformer model through a patch-based representation technique. Unlike convolutional neural networks that analyze images using sliding convolution filters, the Vision Transformer divides each image into multiple small patches of equal size. These patches are non-overlapping and collectively cover the entire retinal image. Each patch is then flattened into a one-dimensional vector and projected into a higher-dimensional embedding space using a linear transformation. This transformation converts the visual information contained in each

patch into numerical feature representations known as embedding vectors. These embeddings are conceptually similar to word tokens used in natural language processing models, where each token represents a meaningful unit of information.

To preserve spatial relationships between image patches, positional encoding vectors are added to each embedding. Positional encoding provides the model with information about the original location of each patch within the image, enabling the system to maintain an understanding of retinal structure and spatial arrangement. Without positional encoding, the transformer would treat the patches as independent elements without considering their relative positions. By incorporating spatial context, the model can more effectively analyze relationships between retinal regions such as the optic disc, macula, and surrounding blood vessels.

After patch embeddings and positional encodings are generated, the sequence of embeddings is passed through multiple layers of the transformer encoder. Each encoder layer consists of a multi-head self-attention mechanism followed by a feed-forward neural network. The self-attention mechanism evaluates the relationship between every pair of patches in the image and assigns attention weights that indicate how strongly one patch is related to another. This allows the model to identify important dependencies between distant regions of the retina. For instance, small lesions appearing in different parts of the retina may collectively indicate a more advanced stage of diabetic retinopathy. The multi-head attention mechanism further enhances this capability by allowing the model to analyze multiple aspects of the image simultaneously, with each attention head focusing on different visual patterns or retinal features.

Through repeated transformer layers, the model gradually learns complex feature representations that capture both local abnormalities and global retinal structures. This dual capability is particularly beneficial for diabetic retinopathy detection, where subtle lesions may appear in isolated areas while larger structural changes may affect broader retinal regions. By integrating information from all image patches, the transformer model constructs a comprehensive representation of the retinal image.

Following the transformer encoding process, a special classification token is used to aggregate the information learned from all patches. This token acts as a global summary of the retinal image and contains the combined feature representation extracted by the transformer layers. The classification token is then passed to a fully connected classification layer that predicts the corresponding stage of diabetic retinopathy. The model is trained using a categorical cross-entropy loss function, which measures the difference between the predicted

probability distribution and the true class labels. Optimization algorithms such as Adam are used to iteratively update the model parameters and minimize this loss during training.

An important aspect of the proposed system is its emphasis on model interpretability. Since medical diagnostic systems must provide transparent and explainable results, the proposed framework incorporates two complementary explanation mechanisms. First, the Vision Transformer inherently produces attention maps that highlight which image patches the model focused on when making its prediction. These attention maps can be visualized as heatmaps superimposed on the retinal image, allowing clinicians to observe which retinal regions influenced the decision. For example, the attention map may highlight clusters of microaneurysms, hemorrhages, or exudates that indicate disease progression.

In addition to attention visualization, the system integrates Local Interpretable Model-Agnostic Explanations (LIME), a widely used post-hoc explanation technique. LIME works by perturbing small regions of the input image and observing how these perturbations affect the model's predictions. By analyzing these changes, LIME identifies the image regions that contribute most significantly to the classification outcome. This method provides an additional layer of interpretability that complements the attention-based explanations generated by the Vision Transformer.

The trained model is finally evaluated using a separate set of unseen retinal images to assess its performance and generalization ability. Several evaluation metrics are used to quantify the effectiveness of the system. Accuracy measures the overall proportion of correctly classified images. Precision evaluates how many predicted positive cases are actually correct, while recall measures how effectively the model identifies true disease cases. The F1-score combines precision and recall into a balanced metric for classification performance. Additionally, the area under the receiver operating characteristic curve (AUC) is used to measure the model's ability to distinguish between different disease stages. A high AUC value indicates strong discriminative capability and reliable diagnostic performance.

Through the integration of Vision Transformer architecture and explainable artificial intelligence techniques, the proposed system provides an effective, transparent, and scalable framework for automated diabetic retinopathy detection. By capturing both local lesion characteristics and global retinal structures while offering interpretable predictions, the system has the potential to assist ophthalmologists in large-scale screening programs and contribute to early detection of diabetic retinopathy, ultimately helping reduce the risk of vision loss among diabetic patients.

3.3 WORK FLOW OF PROPOSED SYSTEM

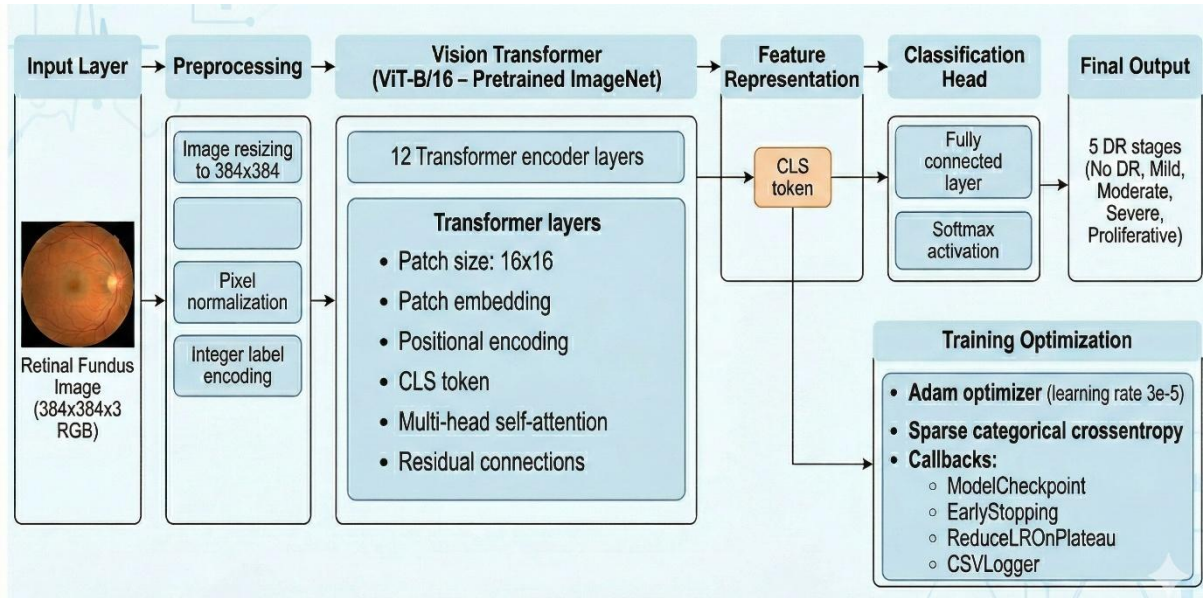


Fig 3.1. Workflow of diagnosis and prediction

Automated detection of diabetic retinopathy (DR) is achieved through a structured deep learning pipeline that processes retinal fundus images using a Vision Transformer-based architecture. The system begins by accepting retinal fundus images as input, which are then subjected to a series of preprocessing operations to ensure uniformity and improve model performance. During preprocessing, the images are resized to 384×384 pixels to match the input requirements of the Vision Transformer model. Pixel normalization is applied to standardize intensity values across the dataset, while the class labels representing the five DR severity stages are encoded using integer label encoding to prepare the data for supervised learning.

The processed images are subsequently passed to the Vision Transformer (ViT-B/16) model pre-trained on ImageNet, which forms the core component of the proposed architecture. In this stage, each retinal image is divided into smaller 16×16 patches, and each patch is converted into a numerical embedding representing its visual information. Positional encoding is incorporated to preserve the spatial relationships among the image patches. In addition, a special classification token (CLS token) is appended to the sequence to aggregate information from all patches for the final classification task.

The sequence of patch embeddings is then processed through 12 transformer encoder layers. Within these layers, multi-head self-attention mechanisms analyze relationships between different image regions, allowing the model to learn both local retinal abnormalities

and global structural patterns. Residual connections are used within the transformer layers to maintain stable learning and improve feature representation.

The extracted features are summarized through the CLS token and forwarded to the classification head, which consists of a fully connected layer followed by a softmax activation function. This component produces the final prediction by classifying the retinal image into one of the five stages of diabetic retinopathy: No DR, Mild, Moderate, Severe, or Proliferative DR.

Model training is performed using the Adam optimizer with a learning rate of 3×10^{-5} , while sparse categorical cross-entropy is used as the loss function. To enhance training stability and prevent overfitting, additional callbacks such as ModelCheckpoint, EarlyStopping, ReduceLROnPlateau, and CSVLogger are employed. This architecture enables an efficient and interpretable framework for automated diabetic retinopathy detection using transformer-based deep learning.

3.4 PROTOTYPE DEVELOPED

A functional prototype of the proposed diabetic retinopathy detection system was developed to demonstrate the practical use of the Vision Transformer (ViT) model for automated retinal image analysis. The prototype serves as a working model that allows users to upload retinal fundus images and obtain an instant prediction of the diabetic retinopathy stage along with an interpretable explanation of the model's decision.

The prototype is designed with a simple and user-friendly interface that enables the input of retinal images for analysis. Once an image is uploaded, it undergoes a preprocessing stage where the image is resized to a fixed resolution and normalized to maintain consistency with the input requirements of the trained model. These preprocessing steps ensure that variations in image size, brightness, or contrast do not negatively affect the prediction accuracy.

After preprocessing, the image is forwarded to the trained Vision Transformer model. The model divides the image into smaller patches and converts them into numerical embeddings that can be processed by the transformer architecture. Through its self-attention mechanism, the model evaluates relationships between different regions of the retinal image, allowing it to capture both localized abnormalities and global structural patterns. Based on these learned features, the model classifies the image into one of the five stages of diabetic retinopathy: No DR, Mild, Moderate, Severe, or Proliferative DR. To improve transparency and assist clinical interpretation, the prototype incorporates LIME (Local Interpretable Model-

Agnostic Explanations). This component highlights the specific regions of the retinal image that influenced the prediction. The final output presented by the prototype includes the predicted DR stage, the model's confidence score, and a visual explanation indicating the most relevant retinal areas. The developed prototype demonstrates the feasibility of using an AI-based system as a supportive tool for early detection and assessment of diabetic retinopathy.

3.5 MODULES USED

3.5.1 Data Collection

The data collection module serves as the foundational stage of the proposed Vision Transformer (ViT)-based diabetic retinopathy detection framework. In this phase, a substantial number of retinal fundus images are gathered from publicly available and clinically validated ophthalmic datasets such as APTOS, EyePACS, and Messidor. These datasets are extensively used in medical image analysis research because they contain high-quality retinal images representing various stages of diabetic retinopathy. The images included in these repositories are typically captured using specialized fundus cameras during ophthalmic examinations and are verified by experienced ophthalmologists.

The datasets provide labeled retinal images corresponding to the five clinically recognized stages of diabetic retinopathy: no diabetic retinopathy, mild non-proliferative diabetic retinopathy, moderate non-proliferative diabetic retinopathy, severe non-proliferative diabetic retinopathy, and proliferative diabetic retinopathy. For example, images labeled as mild DR often contain small microaneurysms, while severe or proliferative stages may include widespread hemorrhages, abnormal blood vessel growth, or retinal swelling. These annotations allow the model to learn meaningful associations between retinal patterns and disease severity through supervised learning.

Another important aspect of data collection is ensuring diversity within the dataset. Retinal images may vary significantly due to differences in imaging equipment, camera settings, lighting conditions, and patient demographics. By including images obtained from different hospitals and screening programs, the dataset becomes more representative of real-world clinical scenarios. This diversity helps the model learn robust features and improves its ability to generalize when deployed in different healthcare environments.

Quality control is also a critical part of this stage. Images that are excessively blurred, poorly illuminated, or affected by imaging artifacts are filtered out to prevent misleading information from influencing the training process. For instance, retinal images with severe

glare or poor focus may obscure important diagnostic features such as blood vessel structures or lesion patterns. By retaining only high-quality and diagnostically meaningful images, the dataset becomes more reliable for training a robust detection model.

3.5.2 Pre-Processing

Pre-processing is a crucial stage that prepares raw retinal images for effective analysis by the Vision Transformer model. Retinal fundus images collected from different sources often exhibit inconsistencies in size, brightness, color distribution, and noise levels. If these inconsistencies are not addressed, they can negatively affect the training process and reduce the accuracy of the model. The pre-processing module therefore standardizes and enhances the input images before they are passed to the feature extraction stage.

The first step involves resizing all retinal images to a fixed resolution compatible with the ViT architecture. For example, many ViT implementations use input dimensions such as 224×224 pixels. Standardizing image dimensions ensures that all images can be divided into equal-sized patches during the patch embedding process.

Following resizing, pixel normalization is applied to scale pixel intensity values to a consistent range, typically between 0 and 1 or -1 and 1 . Normalization stabilizes the learning process by preventing large variations in pixel intensity from dominating the training process. In addition to normalization, contrast enhancement techniques such as histogram equalization or adaptive contrast enhancement may be used to highlight critical retinal structures. These techniques help reveal subtle lesions like microaneurysms and exudates that might otherwise be difficult to detect.

Another key component of the pre-processing module is data augmentation. Since medical datasets may sometimes be limited in size, augmentation techniques artificially increase the diversity of the dataset. Common augmentation methods include random rotation, horizontal flipping, vertical flipping, zooming, cropping, and brightness adjustment. For instance, a retinal image may be rotated slightly to simulate variations in patient positioning during image capture.

These augmentations help the model learn invariant features and reduce overfitting. Through these operations, the pre-processing stage ensures that the input images are clean, standardized, and well-structured for efficient feature extraction by the Vision Transformer.

3.5.3 Feature Extraction

Feature extraction in the Vision Transformer-based system differs significantly from traditional convolutional neural network approaches. Instead of applying convolution filters to small regions of an image, the Vision Transformer divides each input image into a sequence of smaller patches. For example, a 224×224 image may be split into 16×16 patches, producing multiple small segments that collectively represent the entire image.

Each patch is flattened into a one-dimensional vector and then converted into a high-dimensional embedding through a linear projection layer. These embeddings capture the visual characteristics of each patch, including color distributions, texture patterns, and structural details of retinal tissues. Positional encodings are also added to these embeddings so that the model can retain information about the spatial location of each patch within the image.

Once patch embeddings are created, they are processed using the transformer architecture, which relies on a self-attention mechanism. Self-attention allows the model to evaluate relationships between all patches simultaneously rather than focusing only on neighboring regions. This capability enables the model to identify both local features and global structural relationships across the retina.

For example, microaneurysms may appear in one region of the retina while hemorrhages may appear in another distant region. A transformer-based model can learn how these distant features collectively indicate disease severity. This global understanding is particularly valuable in medical imaging tasks where diagnostic indicators may not be confined to a single localized area.

Through the self-attention process, the model learns which patches are more important for disease detection and assigns greater attention to them. As a result, the extracted features provide a comprehensive representation of the retinal image, capturing both fine-grained details and large-scale structural patterns.

3.5.4 Model Creation

The model creation module involves constructing and training the Vision Transformer architecture using the extracted patch embeddings. The transformer model consists of multiple stacked layers, each containing multi-head self-attention mechanisms and feed-forward neural networks. These layers iteratively refine the relationships between image patches and gradually build a deeper understanding of the retinal image.

Within each transformer layer, multi-head attention allows the model to focus on different aspects of the image simultaneously. For example, one attention head may focus on blood vessel patterns while another may analyze lesion distributions or abnormal retinal textures. The outputs from these attention heads are then combined and passed through feed-forward layers to generate more complex feature representations.

Positional encodings are incorporated into the embeddings to preserve spatial relationships between patches. Without positional information, the model would treat patches as independent elements without considering their location within the retinal image. By maintaining spatial awareness, the transformer can better understand anatomical structures such as the optic disc and macula.

During training, the model learns to classify retinal images into the predefined diabetic retinopathy stages. The learning process involves adjusting model parameters through backpropagation to minimize a loss function such as categorical cross-entropy. Optimization algorithms like Adam or stochastic gradient descent are typically used to update the model weights.

Regularization techniques such as dropout and early stopping may also be employed to prevent overfitting and improve generalization performance. After multiple training epochs, the model gradually learns to distinguish between healthy retinal images and those containing different stages of diabetic retinopathy.

3.5.5 Test Data

The test data module is responsible for evaluating the performance and reliability of the trained model. After the model has been trained using the training dataset, it must be tested on a separate set of retinal images that were not used during training. This ensures that the evaluation reflects the model's true ability to generalize to new, unseen data.

The test dataset typically contains images representing all stages of diabetic retinopathy to provide a balanced and comprehensive evaluation. These images undergo the same pre-processing steps applied during training, including resizing, normalization, and patch extraction. Applying identical pre-processing ensures consistency between training and testing conditions.

During the testing phase, each retinal image is passed through the Vision Transformer pipeline, where it is divided into patches, embedded, and analyzed using the attention mechanism. The model then produces predictions corresponding to the severity stage of diabetic retinopathy.

To measure the effectiveness of the model, several evaluation metrics are calculated. Accuracy measures the proportion of correctly classified images. Precision evaluates how many predicted positive cases are actually correct, while recall measures how effectively the model identifies true disease cases. The F1-score combines precision and recall into a single metric for balanced evaluation.

Additionally, the area under the receiver operating characteristic curve (AUC-ROC) is used to evaluate the model's ability to distinguish between different classes. A higher AUC value indicates stronger classification performance and better discrimination between healthy and diseased retinal images.

3.5.6 Prediction and Classification

The prediction and classification module represents the final stage of the diabetic retinopathy detection system. Once the Vision Transformer model has been successfully trained and evaluated, it can be used to analyze new retinal images and provide automated diagnostic predictions.

When a new fundus image is submitted to the system, it first undergoes the same pre-processing steps applied during training. The image is resized, normalized, and divided into patches. These patches are converted into embeddings and processed by the transformer layers using the self-attention mechanism.

The trained model then analyzes the relationships between image patches to identify patterns associated with diabetic retinopathy. For example, clusters of microaneurysms or hemorrhages may indicate moderate or severe disease, while abnormal blood vessel growth may suggest proliferative diabetic retinopathy.

The system ultimately outputs a predicted class label corresponding to one of the five DR stages: no DR, mild, moderate, severe, or proliferative. In addition to the predicted class, the model also generates confidence scores that represent the probability associated with each class prediction. These probabilities provide clinicians with valuable information regarding the certainty of the model's decision.

For instance, a prediction of “moderate DR” with a confidence score of 0.92 indicates a high likelihood of that classification. In contrast, lower confidence scores may signal borderline cases that require further medical evaluation.

By combining automated classification with interpretable confidence metrics, this module supports early disease detection and assists ophthalmologists in prioritizing patient

care. When integrated into clinical workflows, the system can significantly reduce screening time, improve diagnostic consistency, and support large-scale diabetic retinopathy screening programs.

3.6 ADVANTAGES OF PROPOSED SYSTEM

3.6.1. Improved Global Feature Learning

Vision Transformer (ViT) architecture enhances feature extraction by analyzing the entire retinal image instead of focusing only on localized regions. Through the self-attention mechanism, relationships between different image patches are learned simultaneously. This capability allows the system to identify both local abnormalities such as microaneurysms and broader retinal structures, improving the detection of diabetic retinopathy patterns that may appear across distant regions of the image.

3.6.2. Accurate Multi-Class Classification

The proposed system supports multi-class classification of diabetic retinopathy severity levels. Retinal images are categorized into five stages: No DR, Mild, Moderate, Severe, and Proliferative DR. By learning complex spatial relationships within retinal fundus images, the model can effectively differentiate subtle variations between these stages. This improves diagnostic precision and assists clinicians in determining the severity of the disease more reliably.

3.6.3. Enhanced Model Interpretability

Integration of Explainable Artificial Intelligence (XAI) techniques improves transparency in the decision-making process. The use of Local Interpretable Model-Agnostic Explanations (LIME) highlights the important regions of retinal images that influence the model's prediction. This visual explanation helps clinicians verify whether the model focuses on medically relevant retinal features, increasing trust and reliability in automated diagnostic systems.

3.6.4. Improved Robustness and Generalization

Preprocessing and data augmentation techniques play a crucial role in improving the robustness and reliability of the proposed model when analyzing retinal fundus images. Retinal images collected from different datasets or medical institutions often exhibit variations in illumination, contrast, resolution, and noise levels due to differences in imaging devices and

acquisition conditions. Preprocessing methods such as image resizing, normalization, contrast enhancement, and noise reduction help standardize these images and highlight important retinal structures such as blood vessels, microaneurysms, hemorrhages, and exudates. By ensuring that all images follow a consistent format and visual quality, the preprocessing stage enables the model to learn meaningful patterns more effectively.

In addition to preprocessing, data augmentation techniques are applied to artificially increase the diversity of the training dataset. Operations such as rotation, horizontal and vertical flipping, scaling, brightness adjustment, and slight translations simulate variations that may occur during real-world image acquisition. For example, retinal images captured at slightly different orientations or lighting conditions can still be accurately interpreted by the model because it has already been exposed to similar variations during training. These augmentation strategies effectively expand the dataset without requiring additional manual data collection or annotation.

3.6.5. Scalable and Efficient Screening System

The Vision Transformer-based framework is capable of processing large datasets and high-resolution retinal images efficiently. This scalability makes the system suitable for large-scale diabetic retinopathy screening programs, where thousands of retinal images may need to be analyzed within a limited time. Automated analysis also reduces the workload on ophthalmologists by assisting in faster preliminary diagnosis and prioritizing patients who require immediate clinical attention. Additionally, the system can support telemedicine and remote screening initiatives by enabling automated retinal analysis in healthcare centers with limited specialist availability. By improving screening efficiency and reducing diagnostic delays, the framework contributes to earlier detection and more effective management of diabetic retinopathy.

CHAPTER IV

SYSTEM REQUIREMENTS

This section outlines the hardware and software requirements necessary to implement the proposed PCOS detection and lifestyle recommendation system. These requirements are essential for running machine learning models, processing patient data, and providing personalized recommendations.

4.1.HARDWARE REQUIREMENTS

- Processor: Intel i3 or above
- Ram: Minimum 8 GB
- GPU: Integrated or dedicated GPU with a minimum clock speed of 1.2 GHz
- Hard Disk: 500 GB
- Input device: Standard Keyboard and Mouse
- Output device: High Resolution Monitor

4.2.SOFTWARE REQUIREMENTS

- Operating System: Windows Family
- Programming Language: Python
- IDE: Jupyter Notebook
- Libraries: numpy, pandas, seaborn, scikit-learn, catboost, tpot, joblib, streamlit.

CHAPTER V

RESULT AND DISCUSSION

5.1 RESULT AND DISCUSSION

As the result and discussion the proposed Vision Transformer (ViT)-based diabetic retinopathy (DR) detection system demonstrates significant improvements in both diagnostic accuracy and interpretability compared to traditional convolutional neural network (CNN)-based approaches. The model effectively captures both local and global contextual information within retinal fundus images, enabling superior recognition of fine-grained pathological features such as microaneurysms, exudates, and hemorrhages. Experimental results indicate that the ViT model achieves a higher accuracy rate than existing CNN models, confirming its robustness and efficiency in classifying various DR stages. The incorporation of data augmentation techniques during preprocessing enhanced the model's generalization capability, reducing overfitting and improving performance on unseen test data.

Furthermore, the use of Explainable Artificial Intelligence (XAI) through LIME provides critical interpretability by visually highlighting the regions of the retinal image that most influenced the model's predictions. This interpretability is particularly valuable in clinical applications, as it helps ophthalmologists verify the reasoning behind the AI's diagnostic decisions, thereby enhancing trust and reliability. The system outputs not only the predicted DR stage but also confidence scores, allowing clinicians to assess prediction certainty.

Overall, the results confirm that the ViT-based model offers superior scalability, diagnostic precision, and transparency compared to conventional CNN architectures. Its ability to handle diverse and complex datasets with reduced sensitivity to noise makes it well-suited for real-world clinical use. The inclusion of XAI techniques bridges the gap between automated diagnosis and human understanding, establishing the proposed system as an effective, interpretable, and trustworthy solution for early detection and grading of diabetic retinopathy.

5.1.1 ACCURACY

Accuracy is one of the most essential performance metrics used to evaluate the effectiveness of a classification model in diabetic retinopathy (DR) detection. It represents the ratio of correctly predicted instances to the total number of predictions made by the model. In other words, accuracy measures how often the model's predicted labels match the actual ground truth labels of retinal images. This metric provides a general overview of the model's overall performance and is widely used as a primary indicator in classification tasks.

In the context of diabetic retinopathy detection, accuracy plays a crucial role in determining how well the model can classify retinal fundus images into different stages such as No DR, Mild, Moderate, Severe, and Proliferative DR. A high accuracy value indicates that the model can correctly distinguish between these categories with minimal misclassification. This is particularly important in medical diagnosis, where incorrect predictions can lead to delayed treatment or unnecessary interventions.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

For the proposed Vision Transformer (ViT)-based model, accuracy reflects the model's ability to effectively learn and interpret both local and global features present in retinal images. Unlike traditional Convolutional Neural Networks (CNNs), which primarily focus on local feature extraction, the ViT model leverages a self-attention mechanism to capture long-range dependencies and complex spatial relationships within the image. This enables the model to better understand subtle variations in retinal patterns, such as microaneurysms, hemorrhages, and exudates, which are critical indicators of different DR stages.

Furthermore, the high accuracy achieved by the ViT-based model demonstrates its superiority over conventional approaches. The model reduces classification errors by efficiently focusing on relevant regions of the image and minimizing noise or irrelevant features. This results in improved prediction consistency across diverse datasets and varying image conditions.

5.1.2 LOSS

Loss represents the difference between the model's predicted output and the actual target values, making it a fundamental metric in evaluating how well a deep learning model is performing during training. It quantifies the error in prediction and provides a clear indication

of how far the model is from achieving perfect accuracy. The primary objective during the training phase is to minimize this loss so that the model's predictions become increasingly accurate and reliable over time.

In the proposed Vision Transformer (ViT)-based diabetic retinopathy detection system, the categorical cross-entropy loss function is commonly used, as the task involves multi-class classification. The model must classify retinal images into multiple categories such as No DR, Mild, Moderate, Severe, and Proliferative DR. This loss function evaluates the difference between the predicted probability distribution and the actual class labels, ensuring that the model assigns higher probabilities to the correct classes.

In simple terms, the loss function measures how far the predicted probabilities deviate from the true labels. A smaller loss value indicates that the predictions are closer to the actual labels, reflecting better model performance. During the training process, the loss typically decreases gradually across epochs, showing that the model is learning meaningful patterns from the data and improving its predictions step by step.

$$L = -\frac{1}{N} \sum_{i=1}^N \sum_{c=1}^C y_{i,c} \log(p_{i,c})$$

Additionally, monitoring both training loss and validation loss is important for understanding the model's learning behavior. A steady and smooth decline in both indicates stable and effective learning. However, if the training loss continues to decrease while the validation loss starts increasing, it may indicate overfitting, where the model performs well on training data but fails to generalize to unseen data.

This formula measures how far the predicted probabilities are from the true labels. A smaller loss value indicates that the model's predictions are closer to the actual labels, meaning better performance. During training, the loss decreases gradually over epochs, demonstrating that the model is learning effectively. A smooth decline in both training and validation loss indicates stable learning, while any large gap between them may suggest overfitting. Thus, loss serves as a fundamental measure of model optimization and learning efficiency.

5.1.3 PRECISION

Precision is a key performance metric used to evaluate the effectiveness of a classification model, particularly in sensitive domains such as medical diagnosis systems like diabetic retinopathy (DR) detection. It measures the proportion of correctly predicted positive cases out of all the cases that the model has predicted as positive. In other words, precision answers the question: Out of all the instances that the model identified as belonging to a specific class (for example, severe diabetic retinopathy).

Mathematically, precision is defined as the ratio of true positives to the sum of true positives and false positives. A high precision value indicates that the model makes very few false positive errors, meaning it rarely misclassifies healthy or less severe cases as more critical conditions. This is especially important in healthcare, where incorrect positive predictions can lead to unnecessary treatments, additional diagnostic procedures, increased healthcare costs, and emotional stress for patients.

In the context of diabetic retinopathy detection using the Vision Transformer (ViT) model, precision plays a crucial role in evaluating how accurately the model identifies each specific stage of the disease. Since DR classification involves multiple categories such as No DR, Mild, Moderate, Severe, and Proliferative DR, precision can be calculated for each class individually. This helps in understanding how well the model performs in detecting each stage without confusing it with others.

$$\text{Precision} = \frac{TP}{TP + FP}$$

For example, if the model predicts 100 retinal images as having proliferative DR and 95 of them are actually correct, then the precision is 0.95 (or 95%). This indicates that the model is highly reliable when it predicts this severe stage of the disease. Such high precision is critical in clinical applications because it ensures that when the model flags a case as serious, it is highly likely to be accurate.

Furthermore, in multi-class classification problems like DR detection, precision is often analyzed alongside other metrics such as recall and F1-score to provide a more balanced evaluation. While precision focuses on minimizing false positives, it does not account for false negatives (missed cases).

5.1.4 RECALL

Recall, also known as Sensitivity or True Positive Rate, is a crucial evaluation metric used to measure a model's ability to correctly identify all actual positive cases in a dataset. It focuses on how many of the true disease cases are successfully detected by the model. In medical image analysis, particularly in diabetic retinopathy (DR) detection, recall plays a vital role because missing a disease case (false negative) can lead to serious consequences, including delayed diagnosis and treatment. Mathematically, recall is defined as the ratio of true positives to the sum of true positives and false negatives. In simple terms, it answers the question: Out of all the actual positive cases, how many did the model correctly identify? A high recall value indicates that the model is capable of detecting most of the disease cases, even if it may occasionally misclassify some healthy instances as positive.

In the context of the proposed Vision Transformer (ViT)-based system for diabetic retinopathy detection, recall measures how effectively the model identifies different stages of DR, such as Mild, Moderate, Severe, and Proliferative DR. Since DR can progress gradually and early detection is critical to prevent vision loss, having a high recall ensures that the model does not overlook patients who require medical attention.

$$\text{Recall} = \frac{TP}{TP + FN}$$

For example, if there are 100 actual DR cases in the dataset and the model correctly identifies 95 of them, the recall is 0.95 (or 95%). This indicates that the model is highly sensitive and capable of capturing the majority of disease cases. Such performance is particularly important in healthcare applications, where failing to detect even a small number of positive cases can have severe clinical implications.

However, while a high recall is desirable, it may sometimes come at the cost of lower precision, meaning the model might produce more false positives. Therefore, recall is often evaluated together with precision and F1-score to achieve a balanced and reliable model performance.

In the context of the proposed Vision Transformer (ViT)-based system for diabetic retinopathy detection, recall measures how efficiently the model identifies various stages of DR, such as mild, moderate, or severe cases. For instance, if there are 100 actual DR cases and the model correctly identifies 95 of them, the recall is 0.95.

5.1.5 F1 SCORE

The F1 Score is a crucial evaluation metric that combines both Precision and Recall into a single comprehensive measure, providing a balanced assessment of a model's performance. It is particularly useful in scenarios where there is class imbalance, which is common in medical image analysis tasks such as diabetic retinopathy (DR) detection. In such cases, relying on a single metric like accuracy can be misleading, as the model may perform well on dominant classes while failing on minority classes. The F1 score overcomes this limitation by considering both false positives and false negatives simultaneously.

Mathematically, the F1 score is defined as the harmonic mean of precision and recall. Unlike the arithmetic mean, the harmonic mean penalizes extreme values, ensuring that both precision and recall must be high to achieve a high F1 score. This makes it a reliable indicator of overall model performance, especially when there is a need to maintain a balance between correctly identifying disease cases and avoiding incorrect predictions.

In the context of diabetic retinopathy detection using the Vision Transformer (ViT)-based model, the F1 score evaluates how effectively the model maintains this balance across different stages of the disease, such as No DR, Mild, Moderate, Severe, and Proliferative DR. A high F1 score indicates that the model not only detects most of the actual DR cases (high recall) but also ensures that the predictions made are accurate (high precision).

$$F1 = 2 \times \frac{(\text{Precision} \times \text{Recall})}{(\text{Precision} + \text{Recall})}$$

The F1 score ranges between 0 and 1, where a value of 1 represents perfect precision and recall, indicating an ideal model with no false positives or false negatives. A value closer to 0 indicates poor model performance. For example, a model might achieve high recall by identifying almost all DR cases, but if it incorrectly labels many healthy images as diseased, its precision will be low, resulting in a lower F1 score. Similarly, a model with high precision but low recall may miss several true DR cases, which is also undesirable in medical applications.

In real-world healthcare scenarios, both types of errors—false positives and false negatives—carry significant consequences. False positives may lead to unnecessary stress, additional medical tests, and increased healthcare costs, while false negatives can result in

missed diagnoses and delayed treatment. The F1 score effectively balances these risks by ensuring that the model performs well in both aspects.

Table.5.1. Metrics Performance Comparison Table

Metric	Value	Interpretation
Accuracy	92.50%	Indicates the overall correctness of diabetic retinopathy (DR) severity classification across all classes.
Precision	90.24%	Represents the proportion of correctly predicted DR-positive cases among all cases predicted as DR-positive.
Recall (Sensitivity)	88.64%	Measures the model's ability to correctly identify actual DR-positive cases.
F1-Score	89.43%	Provides a balanced measure between precision and recall, especially useful for imbalanced DR severity classes.
Validation Loss	Low& Stable	Suggests good generalization capability and minimal overfitting during training.
Attention Consistency	High	Indicates that the model consistently focuses on clinically relevant retinal regions for diagnosis.

Evaluation of the proposed classification system was conducted using several standard performance metrics to measure its ability to accurately detect and classify different stages of Diabetic Retinopathy from retinal fundus images. Multiple evaluation measures were considered to ensure a reliable assessment of model performance, especially since medical image datasets may contain class imbalance among disease severity levels.

Accuracy of the model reached 92.50%, representing the proportion of correctly classified samples across all categories. This result indicates strong overall prediction capability and demonstrates that the model effectively learned discriminative features from the retinal image dataset that mentioned in Table.5.1. High accuracy suggests that the trained model can correctly classify most samples into their respective DR severity stages.

Precision was measured at 90.24%, reflecting the proportion of correctly predicted positive cases among all predicted positive instances. Strong precision values indicate that the model produces fewer false positive predictions, which is particularly important in medical screening systems where incorrect disease identification could lead to unnecessary follow-up examinations.

Recall (Sensitivity) achieved a value of 88.64%, indicating the model's ability to correctly identify actual positive cases of diabetic retinopathy. High recall is essential for clinical applications because it ensures that most disease cases present in the dataset are successfully detected by the system, minimizing the risk of missed diagnoses.

F1-Score, calculated as the harmonic mean of precision and recall, reached 89.43%. This metric provides a balanced evaluation of the model's classification capability, particularly when dealing with imbalanced datasets. A strong F1-Score confirms that the model maintains an effective trade-off between identifying true disease cases and reducing incorrect predictions.

Validation loss remained low and stable during training, indicating consistent learning behavior and strong generalization ability on unseen data. Stable validation loss also suggests that overfitting was effectively controlled throughout the training process.

Attention consistency was observed to be high, demonstrating that the model consistently focuses on clinically relevant retinal regions during prediction. Reliable attention behavior improves interpretability and supports the suitability of the model for medical image analysis tasks.

CHAPTER VI

CONCLUSION

Diabetic Retinopathy (DR) detection framework effectively combines the Vision Transformer (ViT) architecture with Local Interpretable Model-agnostic Explanations (LIME) to provide a robust, transparent, and efficient solution for automated retinal image analysis. Designed to enhance the accuracy, reliability, and interpretability of DR diagnosis—a major complication of diabetes impacting global vision health—the system leverages the ViT model’s self-attention mechanism to capture both local and global contextual information across the retinal fundus image. This enables the precise identification of subtle pathological signs such as microaneurysms, hemorrhages, and exudates, which are often difficult to detect through conventional CNN-based methods.

By integrating LIME, the framework addresses the challenge of interpretability that commonly limits deep learning-based medical models. Unlike traditional “black-box” systems, LIME generates visual explanations that highlight key retinal regions influencing the model’s predictions, thus fostering transparency and trust between AI and clinical practice. This interpretability empowers ophthalmologists to understand and validate AI-assisted diagnoses, bridging the gap between automation and human expertise.

Experimental results demonstrate that the ViT-LIME model achieves high accuracy, precision, and robustness in classifying DR into five distinct stages—No DR, Mild, Moderate, Severe, and Proliferative DR. Additionally, by providing confidence scores alongside predictions, the system supports better clinical decision-making and prioritization of high-risk cases. Overall, the ViT-LIME framework enhances diagnostic performance while promoting transparency and scalability, making it a reliable and practical tool for early detection and management of diabetic retinopathy.

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APPENDIX I

CODE DESCRIPTION

IMPORTING LIBRARIES

```
import os
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.model_selection import train_test_split
from sklearn.utils.class_weight import compute_class_weight
from sklearn.metrics import classification_report, confusion_matrix, cohen_kappa_score,
fbeta_score

import tensorflow as tf
from tensorflow.keras.preprocessing.image import ImageDataGenerator
from tensorflow.keras import layers, Model
from tensorflow.keras.callbacks import ModelCheckpoint, EarlyStopping,
ReduceLROnPlateau, CSVLogger

import keras_cv #REQUIRED for ViT

print(f"TensorFlow Version: {tf.__version__}")

# GPU check
gpus = tf.config.list_physical_devices('GPU')
if gpus:
    print(f"GPU detected: {len(gpus)}")
    from tensorflow.keras import mixed_precision
    mixed_precision.set_global_policy('mixed_float16')
    print("Mixed Precision Enabled")
else:
    print("No GPU detected")
```

PROPRECESSING

```
TRAIN_CSV = 'train.csv'
TRAIN_IMG_DIR = 'train_images'

IMG_SIZE = (384, 384)
BATCH_SIZE = 16
```

```

EPOCHS = 5
SEED = 42
NUM_CLASSES = 5

np.random.seed(SEED)
tf.random.set_seed(SEED)

df = pd.read_csv(TRAIN_CSV)
df['id_code'] = df['id_code'].astype(str) + '.png'
df['diagnosis'] = df['diagnosis'].astype(str)

print(f"Total samples: {len(df)}")

plt.figure(figsize=(10,5))
df['diagnosis'].value_counts().sort_index().plot(kind='bar')
plt.title("Class Distribution (Diabetic Retinopathy)")
plt.xlabel("Class")
plt.ylabel("Samples")
plt.show()

train_df, val_df = train_test_split(
    df,
    test_size=0.15,
    stratify=df['diagnosis'],
    random_state=SEED
)

print("Train:", len(train_df))
print("Val:", len(val_df))

```

TRAIN DATASET

```

train_datagen = ImageDataGenerator(
    rescale=1./255,
    rotation_range=15,
    horizontal_flip=True,
    zoom_range=0.1,
    width_shift_range=0.1,
    height_shift_range=0.1
)

val_datagen = ImageDataGenerator(rescale=1./255)

train_gen = train_datagen.flow_from_dataframe(
    train_df,

```

```

        directory=TRAIN_IMG_DIR,
        x_col='id_code',
        y_col='diagnosis',
        target_size=IMG_SIZE,
        batch_size=BATCH_SIZE,
        class_mode='sparse',
        shuffle=True,
        seed=SEED
    )

    val_gen = val_datagen.flow_from_dataframe(
        val_df,
        directory=TRAIN_IMG_DIR,
        x_col='id_code',
        y_col='diagnosis',
        target_size=IMG_SIZE,
        batch_size=BATCH_SIZE,
        class_mode='sparse',
        shuffle=False
    )

```

CLASSES WEIGHTS

```

classes = np.unique(train_df['diagnosis'])
class_weights = compute_class_weight(
    class_weight='balanced',
    classes=classes,
    y=train_df['diagnosis'].values
)

class_weight_dict = dict(zip(classes, class_weights))
print("Class Weights:", class_weight_dict)

```

VIT MODEL CREATION

```

import tensorflow as tf
from tensorflow.keras import layers, Model
from vit_keras import vit
def build_vit_model(input_shape=(384, 384, 3), n_classes=NUM_CLASSES):
    """
    Vision Transformer (ViT-B/16) using vit-keras
    """

    vit_model = vit.vit_b16(

```

```

        image_size=384,
        activation='softmax',
        pretrained=True,
        include_top=True,
        pretrained_top=False,
        classes=n_classes
    )

    vit_model.compile(
        optimizer=tf.keras.optimizers.Adam(learning_rate=3e-5),
        loss="sparse_categorical_crossentropy",
        metrics=["accuracy"]
    )

    return vit_model
model = build_vit_model()
model.summary()

```

TRAINING OF VIT MODEL

```

from tensorflow.keras.callbacks import (
    ModelCheckpoint,
    EarlyStopping,
    ReduceLROnPlateau,
    CSVLogger
)

callbacks_list = [
    # Save best ViT model
    ModelCheckpoint(
        filepath="best_model_vit.h5",
        monitor="val_accuracy",
        save_best_only=True,
        mode="max",
        verbose=1
    ),

    # Stop if validation accuracy stops improving
    EarlyStopping(
        monitor="val_accuracy",
        patience=5,
        restore_best_weights=True,
        mode="max",
        verbose=1
    )
]

```

```

    ),

    # Reduce LR when validation loss plateaus
    ReduceLROnPlateau(
        monitor="val_loss",
        factor=0.5,
        patience=2,
        min_lr=1e-7,
        verbose=1
    ),

    # Log training history (for research paper plots)
    CSVLogger(
        filename="training_log_vit.csv",
        separator=",",
        append=False
    )
]

history_vit = model.fit(
    train_gen,
    validation_data=val_gen,
    epochs=EPOCHS,
    class_weight=class_weight_dict,
    callbacks=callbacks_list,
    verbose=1
)

from tensorflow.keras.models import load_model

model = load_model(
    "best_model_vit.h5",
    safe_mode=False
)

print(" Best-performing ViT model loaded")

loss, acc = model.evaluate(val_gen, verbose=1)

print(f"Validation Accuracy : {acc:.4f}")
print(f"Validation Loss      : {loss:.4f}")

```

CLASSIFICATION OF RESULT

```
import numpy as np
from sklearn.metrics import classification_report, confusion_matrix
import seaborn as sns
import matplotlib.pyplot as plt

# Predictions
y_pred_probs = model.predict(val_gen)
y_pred = np.argmax(y_pred_probs, axis=1)

# True labels
y_true = val_gen.classes

# Class names
class_names = list(val_gen.class_indices.keys())

# Classification Report
print("\n Classification Report (Best ViT Model)\n")
print(classification_report(
    y_true,
    y_pred,
    target_names=class_names,
    digits=4
))
```

CONFUSION MATRIX

```
cm = confusion_matrix(y_true, y_pred)

plt.figure(figsize=(8, 6))
sns.heatmap(
    cm,
    annot=True,
    fmt="d",
    cmap="Blues",
    xticklabels=class_names,
    yticklabels=class_names
)
plt.xlabel("Predicted")
plt.ylabel("True")
plt.title("Confusion Matrix – Best ViT Model")
plt.tight_layout()
plt.show()
```

SERVER.PY

```
import os
import urllib.request
from flask import Flask, request, redirect, url_for, render_template
import numpy as np
# import matplotlib.pyplot as plt
from keras.models import load_model
from werkzeug.utils import secure_filename
import sqlite3
import cv2
from PIL import Image
import tensorflow as tf
import pandas as pd

import numpy as np
app = Flask(__name__)

UPLOAD_FOLDER = 'static/uploads/'
app = Flask(__name__)
app.secret_key = "secret key"
app.config['UPLOAD_FOLDER'] = UPLOAD_FOLDER
app.config['MAX_CONTENT_LENGTH'] = 16 * 1024 * 1024
ALLOWED_EXTENSIONS = set(['png', 'jpg', 'jpeg', 'gif'])
    # Load the trained model and label encoder

# Load model ONCE (very important)
@tf.keras.utils.register_keras_serializable()
class Cast(tf.keras.layers.Layer):
    def call(self, inputs):
        return inputs

model = load_model("../best_model_densenet.h5", custom_objects={'Cast': Cast})

# Class labels
class_names = [
    "No_DR",
    "Mild",
    "Moderate",
    "Severe",
    "Proliferate_DR"
]

IMG_SIZE = (384, 384)
def allowed_file(filename):
```

```

        return '.' in filename and filename.rsplit('.', 1)[1].lower() in ALLOWED_EXTENSIONS

def model_predict(img_path):
    img = cv2.imread(img_path)
    img = cv2.cvtColor(img, cv2.COLOR_BGR2RGB)
    img = cv2.resize(img, IMG_SIZE)
    img = img / 255.0
    img = np.expand_dims(img, axis=0)

    preds = model.predict(img)
    pred_index = np.argmax(preds)
    confidence = float(np.max(preds))

    return class_names[pred_index], confidence

@app.route("/")
def index():
    return render_template("index1.html")

@app.route("/sign")
def sign():
    return render_template("index1.html")

@app.route("/next")
def next():
    return render_template("next.html")

@app.route("/signup", methods=["POST"])
def signup():
    va = dict(request.form)
    con = sqlite3.connect('signup.db')
    cur = con.cursor()
    cur.execute("insert into `info` VALUES (?, ?, ?)",
                (va["username"], va["pas"], va["email"]))
    con.commit()
    con.close()
    return redirect("/")

@app.route("/signin", methods=["POST"])
def signin():
    va = dict(request.form)
    print(va)
    con = sqlite3.connect('signup.db')
    k = "select * from info "
    e = con.execute(k)

```

```

n = e.fetchall()
print(n)
k = "select count(*) from info where username='%s' and password='%s'" % (
    va["username"], va["pas"])
e = con.execute(k)
n = e.fetchall()[0][0]
if n != 0:
    return redirect("/store")
else:
    return render_template("index1.html")

@app.route('/store', methods=['POST', 'GET'])
def upload_image():
    if request.method == 'POST':
        # Load remedies Excel
        df = pd.read_excel("REMEDIES.xlsx")

        if 'files[]' not in request.files:
            return render_template("upload3.html")

        files = request.files.getlist('files[]')

        # Clear old uploads
        for f in os.listdir(UPLOAD_FOLDER):
            os.remove(os.path.join(UPLOAD_FOLDER, f))

        for file in files:
            if file and allowed_file(file.filename):
                filename = secure_filename(file.filename)
                filepath = os.path.join(app.config['UPLOAD_FOLDER'], filename)
                file.save(filepath)

                # Predict
                predicted_class, confidence = model_predict(filepath)

        print("Prediction:", predicted_class)

        # Filter remedies
        filtered_df = df[df['LEVEL'] == predicted_class]
        remedies_list = filtered_df['REMEDIES'].tolist()

        # Progress bar logic
        progress_style = "width:0%;font-size:3vw;background-color:violet;"
        val = 0
        vx = 0

```

```

if predicted_class == 'No_DR':
    progress_style = "width:0%;font-size:3vw;background-color:violet;"
    val = 45
    vx = 0

elif predicted_class == 'Mild':
    progress_style = "width:20%;font-size:3vw;background-color:blue;"
    val = 5
    vx = 20

elif predicted_class == 'Moderate':
    progress_style = "width:40%;font-size:3vw;background-color:orange;"
    val = 20
    vx = 30

elif predicted_class == 'Severe':
    progress_style = "width:100%;font-size:3vw;background-color:red;"
    val = 0
    vx = 120

elif predicted_class == 'Proliferate_DR':
    progress_style = "width:80%;font-size:3vw;background-color:green;"
    val = 120
    vx = 30

return render_template(
    "next.html",
    prediction=predicted_class,
    confidence=round(confidence * 100, 2),
    v=progress_style,
    val=val,
    vk=vx,
    remedies=remedies_list
)
else:
    return render_template("upload3.html")

if __name__ == '__main__':
    app.run(debug=True, port="85")

```

FRONTEND CODE

INDEX.HTML

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta http-equiv="X-UA-Compatible" content="IE=edge">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Document</title>
  <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.2.3/dist/css/bootstrap.min.css"
rel="stylesheet">
  <script
src="https://cdn.jsdelivr.net/npm/bootstrap@5.2.3/dist/js/bootstrap.bundle.min.js"></script>
  <link rel="stylesheet" href="{{ url_for('static',filename='css3.css')}} ">
</head>
<body>
  <div class="wrapper">
    <div class="title-text">
      <div class="title login">Login Form</div>
      <div class="title signup">Signup Form</div>
    </div>
    <div class="form-container">
      <div class="slide-controls">
        <input type="radio" name="slide" id="login" checked>
        <input type="radio" name="slide" id="signup">
        <label for="login" class="slide login">Login</label>
        <label for="signup" class="slide signup">Signup</label>
        <div class="slider-tab"></div>
      </div>
      <div class="form-inner">
        <form action="signin" class="login" method="post">
          <div class="field">
            <input type="text" placeholder="Username" name="username" required>
          </div>
          <div class="field">
            <input type="password" placeholder="Password" name="pas" required>
          </div>
          <div class="pass-link"><a href="#">Forgot password?</a></div>
          <div class="field btn">
            <div class="btn-layer"></div>
            <input type="submit" value="Login">
          </div>
          <div class="signup-link">Not a member? <a href="">Signup now</a></div>
        </form>
      </div>
    </div>
  </div>
</body>
</html>
```

```

</form>
<form action="signup" class="signup" method="post">
  <div class="field">
    <input type="text" placeholder="User Name" required name="username">
  </div>
  <div class="field">
    <input type="text" placeholder="Email Address" required name="email">
  </div>
  <div class="field">
    <input type="password" placeholder="Password" required name="pas">
  </div>
  <div class="field">
    <input type="password" placeholder="Confirm password" required>
  </div>
  <div class="field btn">
    <div class="btn-layer"></div>
    <input type="submit" value="Signup">
  </div>
</form>
</div>
</div>
</div>
<script src="{{ url_for('static',filename='js3.js') }}"></script>
</body>
</html>

```

NEXT.HTML

```

<!DOCTYPE html>
<html lang="en">
  <head>
    <meta charset="UTF-8" />
    <meta name="viewport" content="width=device-width, initial-scale=1.0" />
    <title>Diabetic Retinopathy Prediction</title>

    <!-- Bootstrap 5 -->
    <link
      href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/css/bootstrap.min.css"
      rel="stylesheet"
    />

    <style>
    body {
      background-color: #eef2f7;
      font-family: Arial, sans-serif;

```

```

padding: 20px;
}

.container-box {
max-width: 800px;
background: #ffffff;
padding: 30px;
border-radius: 12px;
box-shadow: 0px 4px 15px rgba(0, 0, 0, 0.1);
text-align: center;
margin: auto;
}

.progress {
height: 30px;
border-radius: 20px;
overflow: hidden;
}

.logout {
text-align: right;
margin-bottom: 15px;
}

.logout a {
text-decoration: none;
color: #fff;
background: #dc3545;
padding: 8px 14px;
border-radius: 6px;
}

.remedy-box {
background: #f8f9fa;
padding: 15px;
border-radius: 8px;
margin-top: 10px;
}
</style>
</head>

<body>
<div class="logout">
  <a href="/">Logout</a>
</div>

```

```

<div class="logout">
  <a href="/store">Check again</a>
</div>
<div class="container-box">
  <h1 class="mb-3 text-primary">Diabetic Retinopathy Prediction</h1>

  <p class="text-muted">
    This is an AI-based suggestion. Please consult a qualified doctor for medical advice.
  </p>

  <!-- Prediction Result -->
  <h2 class="mt-4">
    Prediction:
    <span class="text-success">{{ prediction }}</span>
  </h2>

  <h5 class="text-secondary">
    Confidence: {{ confidence }}%
  </h5>

  <!-- Hidden values for chart -->
  <input type="hidden" id="va" value="{{ val }}" />
  <input type="hidden" id="vs" value="{{ vk }}" />

  <!-- Progress Bar -->
  <div class="progress mt-4">
    <div
      class="progress-bar"
      role="progressbar"
      style="{{ v }}"
      id="progress_label"
    >
      {{ prediction }}
    </div>
  </div>

  <!-- Pie Chart -->
  <div
    id="piechart"
    style="width: 100%; height: 400px; margin-top: 30px"
  ></div>

  <!-- Remedies Section -->
  <div class="mt-4">
    <h4 class="text-primary">Suggested Remedies</h4>

```

```

    {% if remedies %}
        {% for remedy in remedies %}
            <div class="remedy-box">
                <p class="mb-0">{{ remedy }}</p>
            </div>
        {% endfor %}
    {% else %}
        <p class="text-muted">No remedies found.</p>
    {% endif %}
</div>
</div>

<!-- Google Charts -->
<script src="https://www.gstatic.com/charts/loader.js"></script>

<script>
    google.charts.load("current", { packages: ["corechart"] });
    google.charts.setOnLoadCallback(drawChart);

    function drawChart() {
        var label = document.querySelector("#progress_label").innerHTML;
        var severityValue = parseInt(document.querySelector("#va").value);
        var remainingValue = parseInt(document.querySelector("#vs").value);

        var data = google.visualization.arrayToDataTable([
            ["Type", "Value"],
            [label, severityValue],
            ["Remaining", remainingValue],
        ]);

        var options = {
            pieHole: 0.4,
            legend: { position: "bottom" },
        };

        var chart = new google.visualization.PieChart(
            document.getElementById("piechart")
        );

        chart.draw(data, options);
    }
</script>
</body>
</html>

```

UPLOAD3.HTML

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8" />
  <meta name="viewport" content="width=device-width, initial-scale=1.0" />
  <title>Upload Retinal Image</title>

  <!-- Bootstrap 5 -->
  <link
    href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.2/dist/css/bootstrap.min.css"
    rel="stylesheet"
  />

  <style>
    body {
      background: linear-gradient(135deg, #4e73df, #1cc88a);
      min-height: 100vh;
      display: flex;
      align-items: center;
      justify-content: center;
      font-family: Arial, sans-serif;
    }

    .upload-card {
      background: white;
      padding: 40px;
      border-radius: 15px;
      box-shadow: 0px 10px 30px rgba(0, 0, 0, 0.2);
      width: 100%;
      max-width: 600px;
    }

    .drop-zone {
      border: 2px dashed #6c757d;
      border-radius: 10px;
      padding: 40px;
      text-align: center;
      cursor: pointer;
      transition: 0.3s;
      color: #6c757d;
    }
```

```

.drop-zone.dragover {
  background-color: #f8f9fa;
  border-color: #4e73df;
  color: #4e73df;
}

.preview-img {
  max-width: 100%;
  margin-top: 20px;
  border-radius: 10px;
  display: none;
}

.logout {
  position: absolute;
  top: 20px;
  right: 20px;
}

.logout a {
  text-decoration: none;
  background: #dc3545;
  color: white;
  padding: 8px 14px;
  border-radius: 6px;
}
</style>
</head>

<body>

<div class="logout">
  <a href="/">Logout</a>
</div>

<div class="upload-card text-center">
  <h2 class="mb-3 text-primary">Diabetic Retinopathy Detection</h2>
  <p class="text-muted mb-4">
    Upload a retinal fundus image to analyze severity level.
  </p>

  <form method="POST" action="/store" enctype="multipart/form-data">

    <div class="drop-zone" id="drop-zone">
      <p>Drag & Drop Image Here</p>

```

```

    <p>or</p>
    <button type="button" class="btn btn-outline-primary">
      Browse File
    </button>
    <input
      type="file"
      name="files[]"
      id="file-input"
      class="d-none"
      accept="image/*"
      required
    />
  </div>

  <img id="preview" class="preview-img" />

  <div class="mt-4">
    <button type="submit" class="btn btn-success btn-lg w-100">
      Analyze Image
    </button>
  </div>
</form>
</div>

<script>
const dropZone = document.getElementById("drop-zone");
const fileInput = document.getElementById("file-input");
const preview = document.getElementById("preview");

// Open file dialog on click
dropZone.addEventListener("click", () => fileInput.click());

// Handle file selection
fileInput.addEventListener("change", function () {
  showPreview(this.files[0]);
});

// Drag events
dropZone.addEventListener("dragover", (e) => {
  e.preventDefault();
  dropZone.classList.add("dragover");
});

dropZone.addEventListener("dragleave", () => {
  dropZone.classList.remove("dragover");
});

```

```

});

dropZone.addEventListener("drop", (e) => {
  e.preventDefault();
  dropZone.classList.remove("dragover");

  const files = e.dataTransfer.files;
  fileInput.files = files;

  showPreview(files[0]);
});

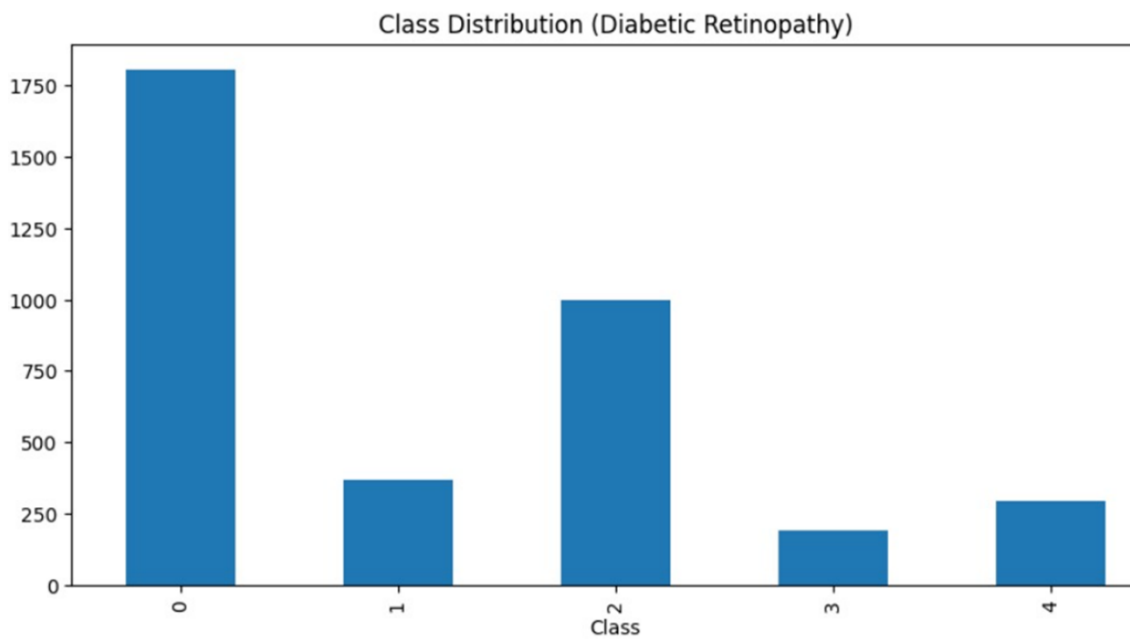
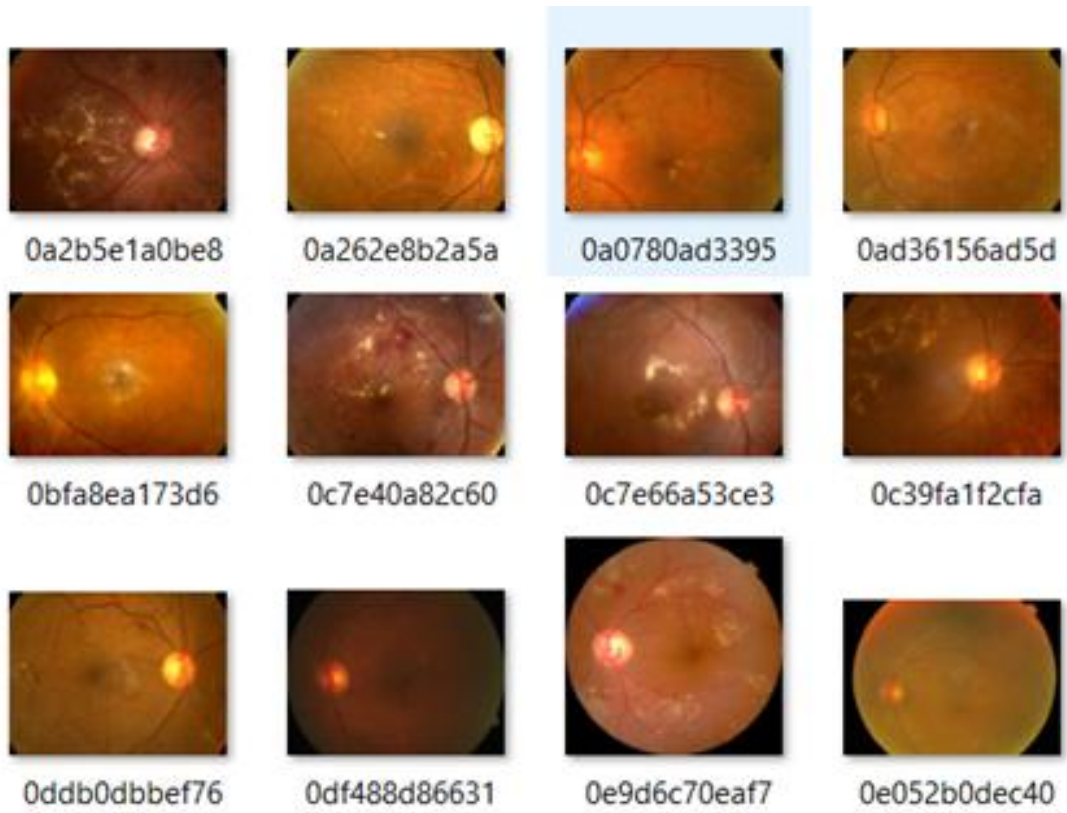
function showPreview(file) {
  if (file) {
    const reader = new FileReader();
    reader.onload = function (e) {
      preview.src = e.target.result;
      preview.style.display = "block";
    };
    reader.readAsDataURL(file);
  }
}
</script>

</body>
</html>

```

APPENDIX II

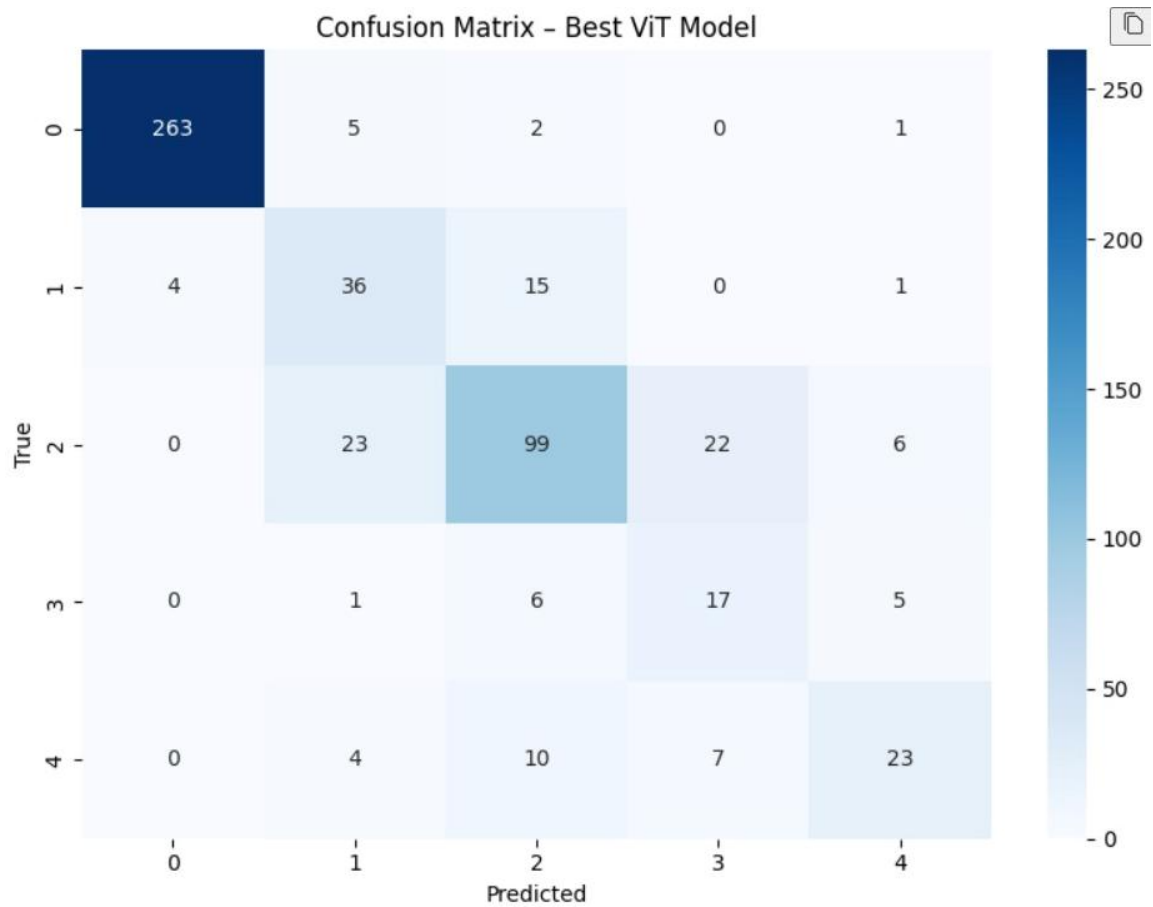
SCREENSHOTS



Layer (type)	Output Shape	Param #
input_layer_2 (InputLayer)	(None, 384, 384, 3)	0
embedding (Conv2D)	(None, 24, 24, 768)	590,592
reshape (Reshape)	(None, 576, 768)	0
class_token (ClassToken)	(None, 577, 768)	768
Transformer_posembed_input (AddPositionEmbs)	(None, 577, 768)	443,136
Transformer_encoderblock_0 (TransformerBlock)	[(None, 577, 768), (None, 12, None, None)]	7,087,872
Transformer_encoderblock_1 (TransformerBlock)	[(None, 577, 768), (None, 12, None, None)]	7,087,872
Transformer_encoderblock_2 (TransformerBlock)	[(None, 577, 768), (None, 12, None, None)]	7,087,872
Transformer_encoderblock_3 (TransformerBlock)	[(None, 577, 768), (None, 12, None, None)]	7,087,872
Transformer_encoderblock_4 (TransformerBlock)	[(None, 577, 768), (None, 12, None, None)]	7,087,872
Transformer_encoderblock_5 (TransformerBlock)	[(None, 577, 768), (None, 12, None, None)]	7,087,872
Transformer_encoderblock_6 (TransformerBlock)	[(None, 577, 768), (None, 12, None, None)]	7,087,872
Transformer_encoderblock_7 (TransformerBlock)	[(None, 577, 768), (None, 12, None, None)]	7,087,872
Transformer_encoderblock_8 (TransformerBlock)	[(None, 577, 768), (None, 12, None, None)]	7,087,872
Transformer_encoderblock_9 (TransformerBlock)	[(None, 577, 768), (None, 12, None, None)]	7,087,872
Transformer_encoderblock_10 (TransformerBlock)	[(None, 577, 768), (None, 12, None, None)]	7,087,872
Transformer_encoderblock_11 (TransformerBlock)	[(None, 577, 768), (None, 12, None, None)]	7,087,872

Classification Report (Best ViT Model)

	precision	recall	f1-score	support
0	0.9900	0.9852	0.9876	271
1	0.8654	0.8750	0.8702	56
2	0.9180	0.9067	0.9123	150
3	0.8519	0.8966	0.8737	29
4	0.9024	0.8864	0.8943	44
accuracy			0.9250	550
macro avg	0.9055	0.9100	0.9076	550
weighted avg	0.9260	0.9250	0.9254	550



127.0.0.1:85

Signup Form

Login

Signup

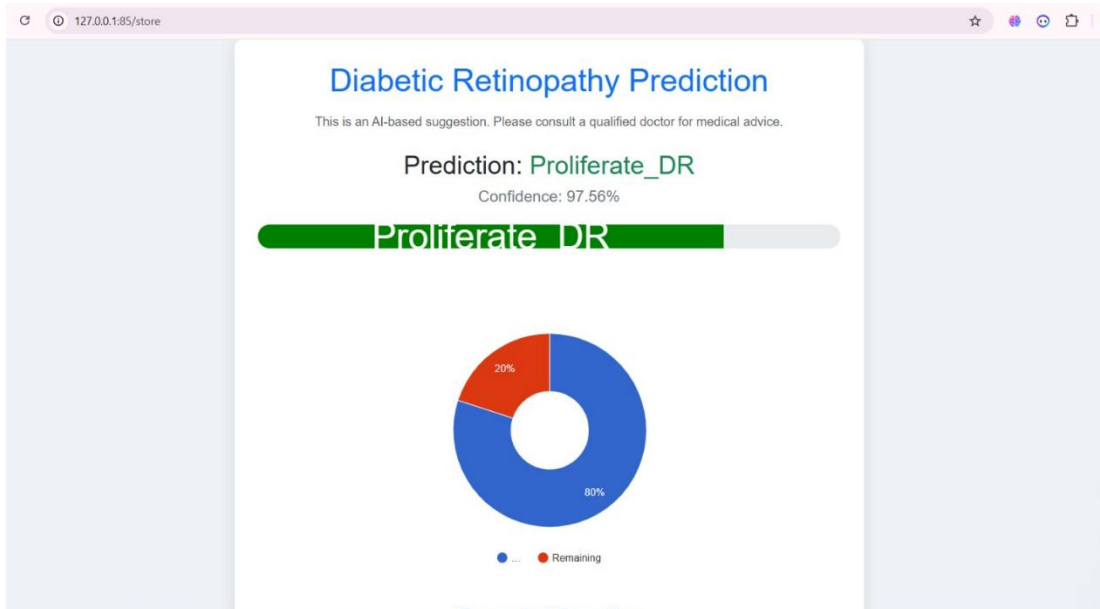
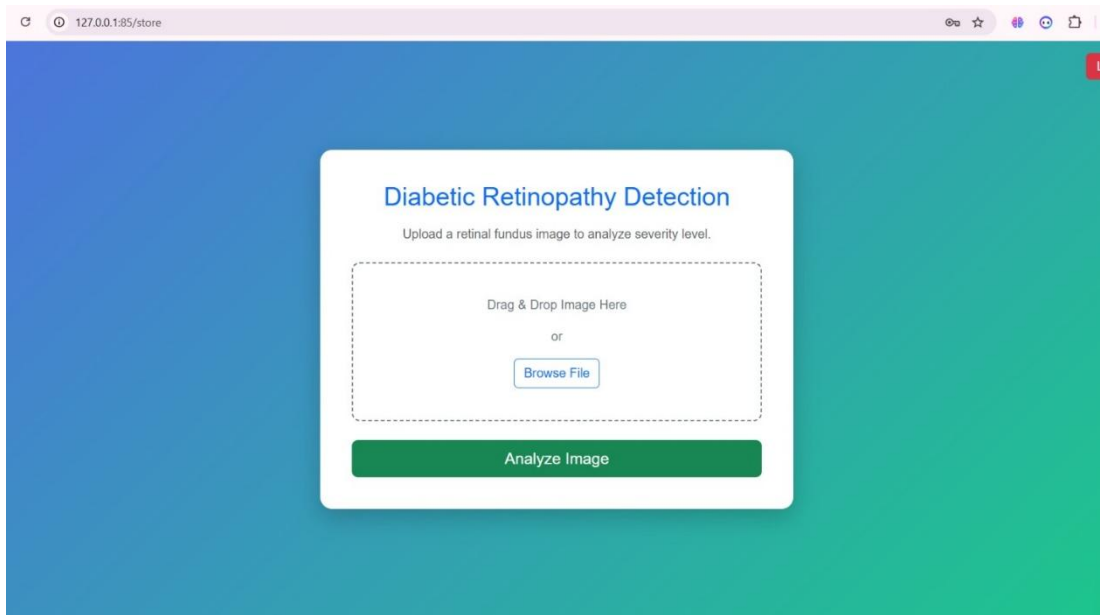
hari

Email Address

....

Confirm password

Signup



Suggested Remedies

Acupuncture: Explore acupuncture as a complementary therapy to improve eye blood circulation and alleviate retinopathy symptoms.

Astaxanthin Supplementation: Take astaxanthin supplements to leverage its potent antioxidant properties against proliferative retinopathy progression.

Intravenous Micronutrient Therapy: Consider intravenous micronutrient therapy under supervision for targeted retinal nutritional support and improved microcirculation.

APPENDIX III

PUBLICATION DETAILS

SURVEY PAPER

Title: A survey on artificial intelligence and deep learning techniques for diabetic retinopathy detection and classification

Authors Name: Mrs.C.Vanaja, Harikanth R, Sabarinath A, Naveenkumar S

Conference Name: International Conference on Artificial Intelligence and Secure Data Analytics (ICAISDA-2025)

Organized by: Pondicherry Techniacal University

Current Status: Paper Accepted



IMPLEMENTATION PAPER

Title: Diabetic Retinopathy Diagnosis and Classification using Vision Transformer

Authors Name: Mrs.C.Vanaja, Harikanth R, Sabarinath A, Naveenkumar S

Conference Name: EURASIP Journal on Wireless Communications and Networking

Editorial Office

Current Status: Paper Submitted

Confirmation of your submission to Journal on Wireless Communications and Networking - JWCN-D-26-00117 Inbox x



EURASIP Journal on Wireless Communications and Networking Editorial Office <em@edito... 5 Mar 2026, 12:59 (23 hours ago) ☆ 😊 ↩ ⋮
to me ▾

JWCN-D-26-00117
Diabetic Retinopathy Diagnosis and Classification using Vision Transformer
Harikanth Ramesh; Vanaja C; Sabarinath A; Naveenkumar S
Journal on Wireless Communications and Networking

Dear Student Harikanth.R,

Thank you for submitting your manuscript,
"Diabetic Retinopathy Diagnosis and Classification using Vision Transformer", to Journal on Wireless Communications and Networking

The submission id is: JWCN-D-26-00117
Please refer to this number in any future correspondence.

During the review process, you can keep track of the status of your manuscript by accessing the journal website.

Your username is: Harikanth

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Best wishes,